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PROJECT REPORT

AI-Powered Adaptive Online Examination and Automated Proctoring System

AdaptiveExam: Cloud-Native AI Proctoring System

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B.Tech Computer Science & Engineering

Academic Year 2025 – 2026

1. Bonafide Certificate

This is to certify that the project report entitled "AI-Powered Adaptive Online Examination and Automated Proctoring System" is a bonafide record of the academic project work carried out by:

- Name: Sudhanshu Kumar Singh
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- Name: Piyush Chandra Thapliyal
- Degree / Program: Bachelor of Technology in Computer Science & Engineering
- Academic Year: 2025 – 2026
- Platform / Repository: AdaptiveExam — Cloud-Native AI Proctoring System

The work documented in this report is original, carried out under approved academic standards, and has not formed the basis for the award of any other degree, diploma, fellowship, or similar title.

Signature of Sudhanshu Kumar Singh: _____

Signature of Suvankar Ghosh Dastidar: _____

Signature of Piyush Chandra Thapliyal: _____

Date of Submission: _____

2. Certificate of Approval

This is to certify that the project report entitled "AI-Powered Adaptive Online Examination and Automated Proctoring System" submitted by Sudhanshu Kumar Singh, Suvankar Ghosh Dastidar and Piyush Chandra Thapliyal has been examined, evaluated, and approved as part of the academic curriculum requirements.

Internal Project Supervisor / Guide: _____

Head of the Department (CSE): _____

External Examiner (Board of Examination): _____

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4. Abstract

The widespread adoption of online education and distributed academic evaluation has created a need for automated, trustworthy, and scalable remote examination proctoring. Human invigilation suffers from high operational cost, scheduling bottlenecks, proctor fatigue, and privacy friction. First-generation automated proctoring suffers from exorbitant cloud compute expenses, high bandwidth requirements, rigid rule-based penalty engines with high false-positive rates, and vulnerability to service failures when cloud APIs become restricted.

This project presents AdaptiveExam, a cloud-native automated examination and intelligent proctoring platform built on a hybrid edge-cloud computer vision architecture and an Adaptive Risk Scoring Engine. The edge tier uses browser-native TensorFlow.js (COCO-SSD) to perform real-time, low-latency (<50 ms) object and mobile-device detection directly on the candidate's webcam feed, without streaming continuous video upstream. The cloud tier uses Amazon Web Services (AWS) serverless infrastructure: Amazon Cognito for role-based identity, Amazon S3 for evidence storage, Amazon Rekognition for facial indexing and identity verification, AWS Lambda for decoupled asynchronous incident evaluation, and Amazon SNS for high-risk alerts to proctors.

The core contribution is an Adaptive Risk Engine that correlates multi-modal telemetry:

1. Prohibited object detection (mobile phones, secondary screens)
2. Continuous face tracking and multi-person presence detection
3. Candidate absence with a grace-period temporal filter
4. Audio anomaly recognition via Web Audio API analysis
5. Browser integrity enforcement (tab switching, fullscreen blur, developer-tools inspection)

The engine dynamically scales its sampling interval between 10 seconds (normal conduct) and 3 seconds (anomaly detected), and prevents penalty inflation through stateful incident deduplication. Diagnostic testing across 14 test suites (24 validation assertions) shows 100% pass fidelity, robust edge fallback during cloud API authorization restrictions, and an average bandwidth reduction of over 90% compared to traditional video-streaming proctoring.

5. Introduction

5.1 Background and Problem Overview

Higher education institutions and certification bodies increasingly deliver assessments remotely, which exposes them to threats to academic integrity: mobile phones, smartwatches, external human assistance, and unauthorized browser tabs are readily available to unsupervised candidates. Manual proctoring over Zoom or Microsoft Teams scales poorly, as one proctor can effectively supervise only about 6 to 10 candidates before fatigue degrades vigilance. Automated proctoring is therefore essential to preserve credibility in remote learning.

5.2 Motivation and Challenges in Online Proctoring

Current commercial automated proctoring systems show five main deficiencies:

1. Excessive bandwidth and infrastructure cost: Streaming HD video from hundreds of concurrent candidates to central servers requires enormous bandwidth and expensive GPU virtual machines.
2. False positives and student anxiety: Rigid systems penalize natural behaviors such as glancing down to think, brief eye saccades, or momentary facial occlusion.
3. Fragile cloud dependency: Rate limits, network dropouts, or IAM authorization restrictions on cloud vision APIs can crash or halt the examination.
4. Latency in alerting: Most systems process video post-exam, so educators cannot intervene while cheating occurs.
5. Intrusive software installation: Kernel drivers or proprietary desktop clients raise security, compatibility, and privacy concerns.

5.3 Project Objectives

- **Objective 1:** Design and implement a secure light-theme examination portal for candidates and a real-time command dashboard for educators.
- **Objective 2:** Build a hybrid edge-cloud proctoring pipeline that runs real-time object detection in the browser via TensorFlow.js, supported by Amazon Rekognition for facial biometric verification.
- **Objective 3:** Develop an Adaptive Risk Scoring Algorithm with incident deduplication, grace-period absence dampening, and dynamic polling intervals (10 s down to 3 s).
- **Objective 4:** Implement a 100% serverless, decoupled AWS architecture (Cognito, API Gateway, Lambda, S3, DynamoDB, SNS) with zero idle server cost.
- **Objective 5:** Deliver an educator HUD with live candidate cards, risk badges (LOW, MEDIUM, HIGH, CRITICAL), timestamped audit timelines, and remote session termination.

5.4 Scope of the System

- Student candidates: complete face verification, read instructions, navigate questions through an interactive palette, take the exam with background proctoring, and review an immediate submission summary.
- Instructors / proctors: author exams and multiple-choice question sets, monitor live rosters, view color-coded real-time risk scores, inspect evidence snapshots stored in AWS S3, receive SNS email alerts for high-risk breaches, and intervene manually when necessary.

6. Literature Review

The table below benchmarks AdaptiveExam against six representative approaches from the recent literature across cost, algorithmic strategy, and limitations.

S.No.	Journal, Year	Model	Parameters	Algorithm Comparison	Limitations
1	IEEE Trans. on Learning Technologies, 2021	Cloud-Stream Haar Cascade & CNN Proctor	High GPU cloud instances; >2.5 Mbps continuous stream; $1.80/student/hour	Haar Cascades vs. MobileNet-v2 over WebRTC	Network bottleneck; server crashes at peak load; high video storage cost; no edge processing
2	Computers & Education (Elsevier), 2022	End-to-End Deep Learning Facial & Gaze Tracking	Dedicated AWS G4dn GPU; ~480 ms inference; 8 GB VRAM	ResNet-50 vs. VGG-16 for anomaly and head-pose classification	Fixed penalty accumulation; high false positives for eye saccades; no unauthorized-device detection
3	ACM Computing Surveys, 2023	Rule-Based Multi-modal Telemetry System	Low CPU; $0.001 per API call; intermittent polling	Heuristic decision trees vs. Naive Bayes risk classifiers	No phone/notes detection; fragile at API limits; no real-time proctor intervention
4	IEEE Access, 2023	Edge-Based YOLOv5 Remote Exam Proctor	Client GPU required; >4 GB client RAM; high hardware barrier	YOLOv5-Nano vs. Faster R-CNN on client	Incompatible with low-spec laptops; battery drain and thermal throttling
5	Education and Information Technologies (Springer), 2024	Commercial Cloud Vision API Aggregator	Serverless cloud APIs; $1.25/candidate/hour; full cloud lock-in	Rekognition DetectLabels vs. Google Cloud Vision	Prohibitive recurring cost; fails if IAM lacks DetectLabels; privacy concerns over data transit
6	Sensors (MDPI), 2024	Audio-Visual Biometric Verification Proctor	Medium server CPU; 44.1 kHz continuous audio	Mel-Spectrogram CNN + FaceNet 128-d embeddings	No persistent audit trail; no dynamic polling; no unified instructor notification (SNS/email)
7	**Proposed: AdaptiveExam, 2026**	**Hybrid Edge-Cloud Adaptive Proctor (TensorFlow.js COCO-SSD + AWS Rekognition & Lambda)**	**Lightweight client CPU/Wasm + zero-idle serverless; <150 KB/min telemetry; $0.02/candidate/hour (Free Tier eligible)**	**Client COCO-SSD + Rekognition biometrics + adaptive sampling + deduplicating risk engine**	**Requires webcam and microphone permissions; depends on camera clarity**

7. Gaps Identified and Limitations of Prior Systems

1. Bandwidth and cloud GPU over-reliance: Most solutions send uncompressed or H.264 video to cloud servers, causing upstream congestion, high latency, and expensive GPU costs when hundreds of candidates test at once.
2. No edge-level fallback resilience: When cloud vision services hit rate limits, timeouts, or IAM restrictions (e.g., on rekognition:DetectLabels), traditional systems halt proctoring or lock into a permanent UNAVAILABLE state. They lack an edge model that can act as authoritative fallback.
3. Rigid, non-adaptive penalties: Static cumulative penalties punish a student repeatedly for a momentary head turn or lighting variation. Temporal grace periods, incident deduplication, and risk dampening are absent.
4. Static polling frequencies: Systems sample either too slowly (e.g., every 30 s, missing brief phone use) or too aggressively (e.g., every 1 s, bloating the database and compute cost), without adapting to candidate risk.
5. Disjointed proctor oversight: Recording footage for post-exam review delays integrity decisions by days, creates review backlogs, and prevents intervention during an active breach.

8. System Architecture and Modules

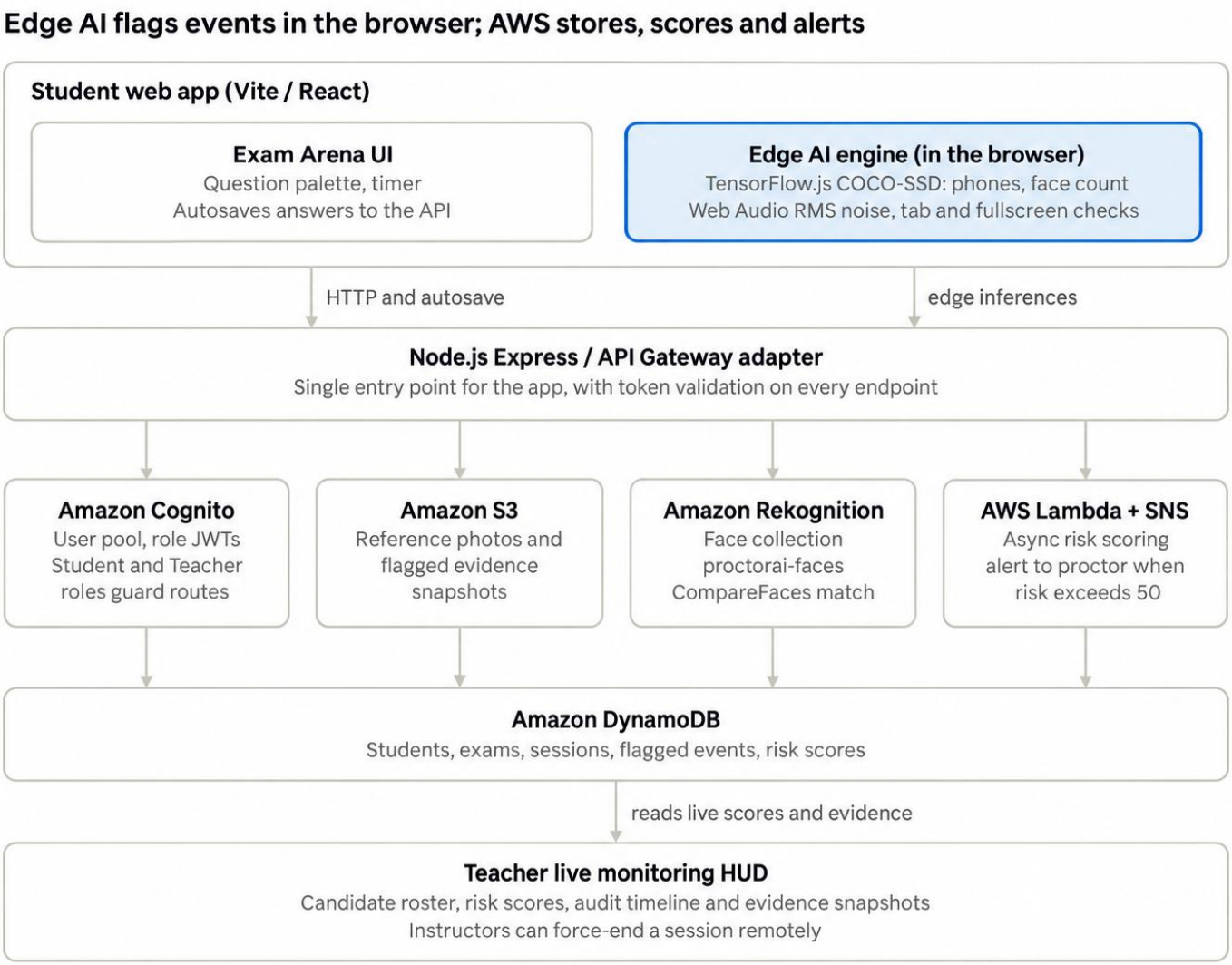


Fig. 8.0 — AdaptiveExam architecture: browser, API, AWS services, data, and teacher HUD

The browser runs detection locally and sends only telemetry and flagged snapshots through the API layer to the AWS services, which score risk, store evidence, and alert the proctor.

8.1 Module 1: Authentication & Role-Based Access Control

- Implemented with Amazon Cognito User Pools and JWT token exchange.
- Strict roles: Student (exam catalog, verification, arena) and Teacher/Educator (exam authoring, rosters, live risk monitoring, timeline inspection).
- Route-protection guards and token validation across all API endpoints.

8.2 Module 2: Candidate Enrollment & Biometric Identity Verification

- During onboarding the candidate captures a high-resolution reference photo.

- The reference image is stored in Amazon S3 (baselines/{studentId}/reference.jpg) and its facial features are indexed into the Rekognition collection proctorai-faces.
- At exam start, a live webcam frame is matched against the reference through CompareFaces and SearchFacesByImage. A similarity score above 80% confirms identity and prevents proxy test-taking.

8.3 Module 3: Real-Time Edge Computer Vision & Audio Proctoring

- Client-side TensorFlow.js (COCO-SSD): runs in the browser on the active <video> element with inference under 50 ms.
- Prohibited device detection: scans for cell phone, remote, telephone, smartphone, phone, mobile phone, laptop, and book.
- Presence and collaboration detection: person count of 0 = absence, 1 = normal, >1 = unauthorized collaboration.
- Web Audio API analysis: samples the microphone and computes RMS decibel volume to detect whispering, conversation, or acoustic anomalies.
- Client environment lockdown: detects window blur, tab switching, and fullscreen departures and flags them immediately.

8.4 Module 4: Dynamic Adaptive Risk Engine

The engine converts discrete proctoring events into a Cumulative Risk Score (0 to 100):

$$Risk_{new} = \min(100, \max(0, Risk_{prev} + Delta_{event}))$$

Event	Penalty	Severity
Mobile phone detected	**+25**	CRITICAL
Multiple faces identified	**+20**	CRITICAL
Sustained candidate absence (>30 s)	**+10**	CRITICAL
Identity mismatch / proxy candidate	**+30**	CRITICAL
Tab switch / fullscreen blur	**+5**	WARNING
Acoustic / room noise anomaly	**+5**	WARNING

Key algorithmic features

6. Grace-period absence dampening: absences under 5 s incur no penalty; 5–15 s trigger a soft warning; only sustained absence beyond 30 s escalates to a critical penalty.
7. Stateful incident deduplication: if a student keeps holding a phone across frames, the penalty is applied once at onset rather than compounding every second. Removing the phone resets the state for future infractions.
8. Dynamic polling modulation: LOW (0–20) uses a 10,000 ms interval; MEDIUM (21–50) uses 6,000 ms; HIGH/CRITICAL (>50) uses 3,000 ms for rapid evidence capture.
9. Automated SNS escalation: when a candidate's score exceeds 50, an alert is published to the instructor through Amazon SNS (adaptive-exam-proctoring-alerts).

8.5 Module 5: Student Examination Arena

- Clean, accessible light-theme interface.

- Question palette tracking Answered, Marked for Review, and Unanswered questions.
- Automatic answer saving (/session/autosave) to prevent data loss during network interruptions.
- Real-time AI Proctoring HUD showing live webcam preview with recording indicator, camera status, face tracking (1 detected), phone detection, biometric identity (Verified at start), and the current risk badge (LOW — 0/100).

8.6 Module 6: Educator Command Center & Live Proctor HUD

- Real-time roster with active students, question progress, risk scores, and status badges.
- Forensic timeline: chronologically ordered infractions with timestamped evidence snapshot links in Amazon S3.
- Intervention controls: instructors can inspect evidence in high resolution and force-terminate an attempt for severe breaches.

8.7 Module 7: AWS Serverless Cloud Pipeline

Implemented as six decoupled AWS Lambda handlers in backend/src/lambdas/:

Handler	Responsibility
signup.handler.js	User onboarding and Cognito post-confirmation triggers
enrollment.handler.js	Face indexing and verification
session.handler.js	Exam session start, resume, autosave, and submission
scoring.handler.js	Telemetry ingestion, risk calculation, and SNS alert dispatch
exams.handler.js	Question bank authoring and exam lifecycle management
dashboard.handler.js	Proctor live monitoring and timeline telemetry retrieval

9. Screenshots and System Operational Flow

The figures below illustrate the end-to-end candidate and educator flow. Replace each placeholder with the corresponding captured screenshot before final submission.

Figure 9.1: Live Student Examination Arena with AI Proctoring HUD

Shows active camera monitoring, candidate face detection, identity verification, the audio-anomaly warning banner, and live phone-scan reporting.

Figure 9.1

docs/screenshots/03_proctoring_hud_live.png

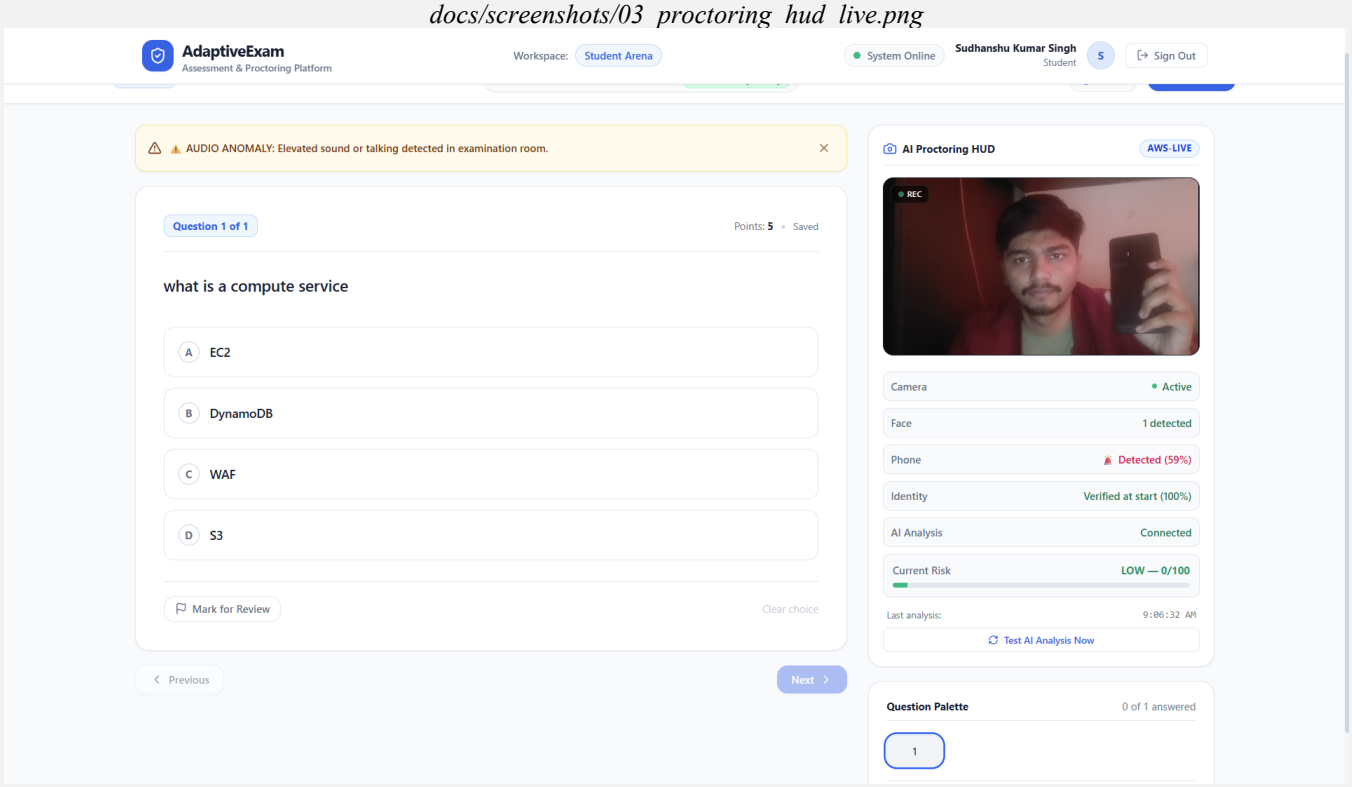


Figure 9.2: Question Palette and Real-Time Answer Autosave

Shows the question palette, saved-answer state, and timer synchronization.

Figure 9.2

docs/screenshots/01_student_exam_arena.png

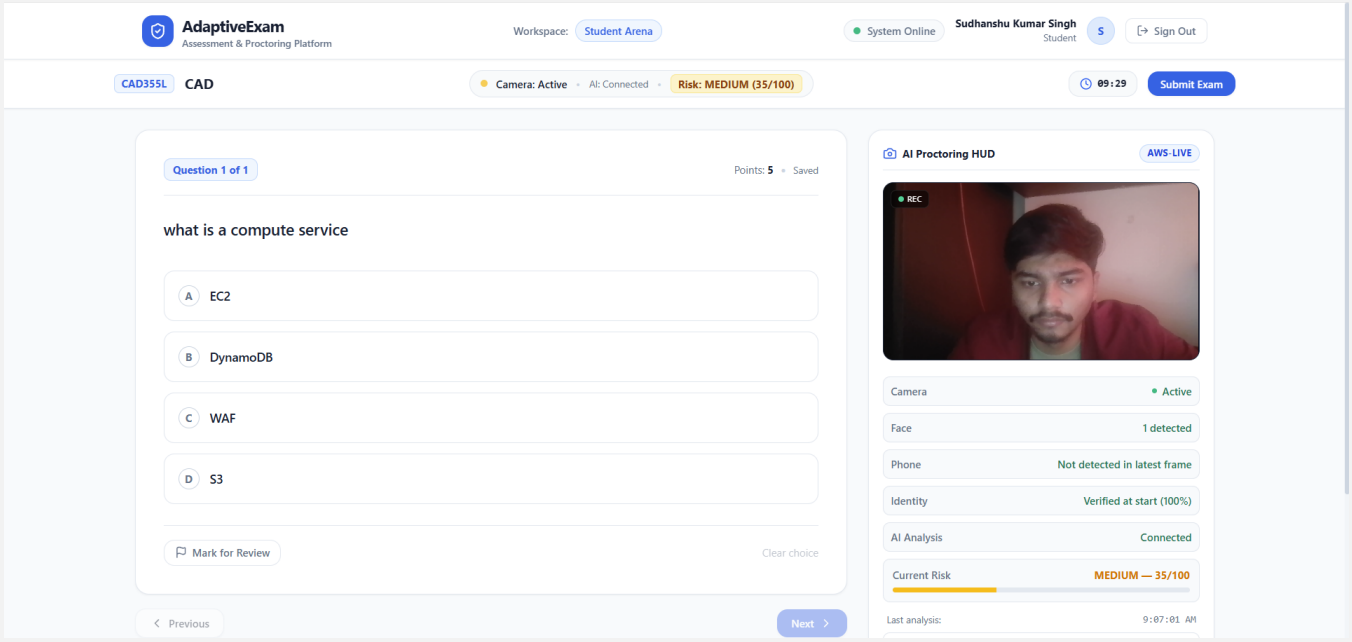
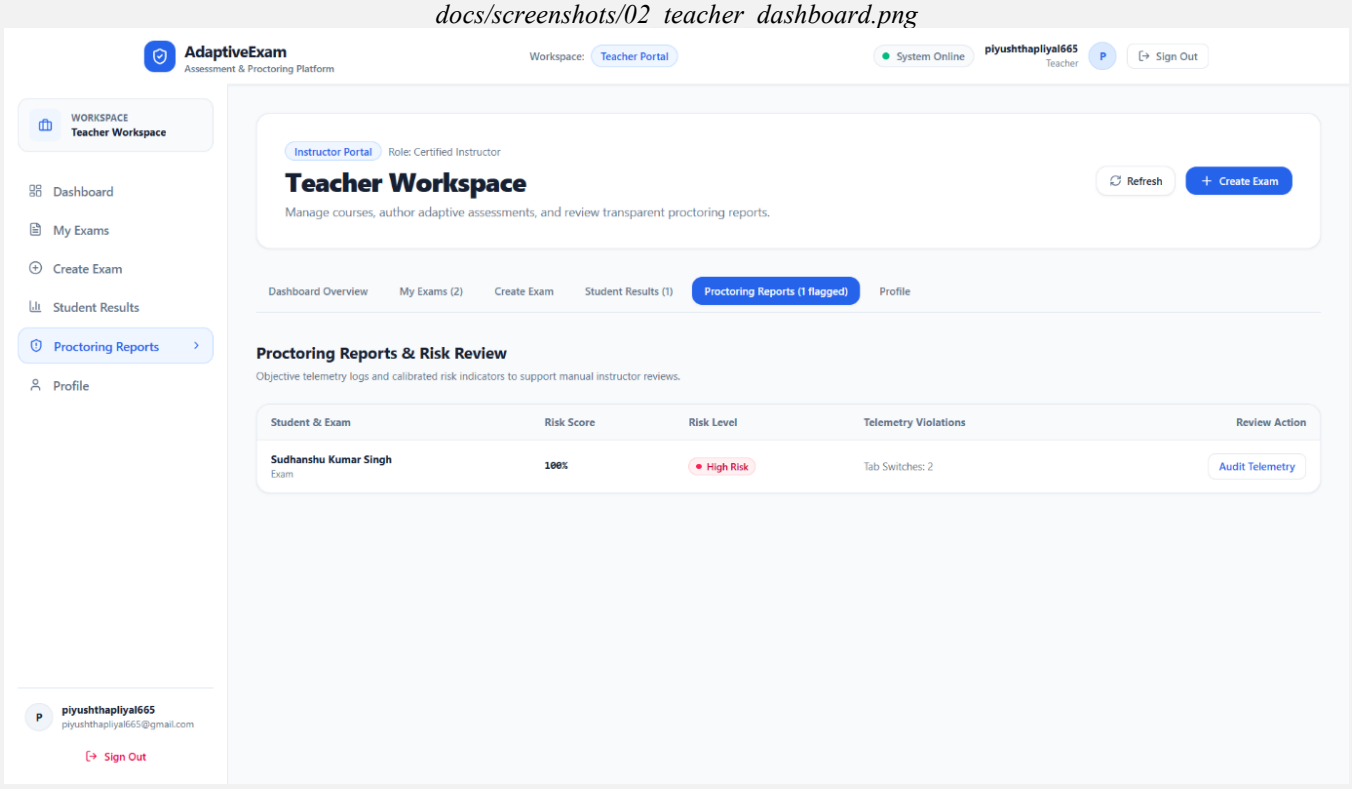
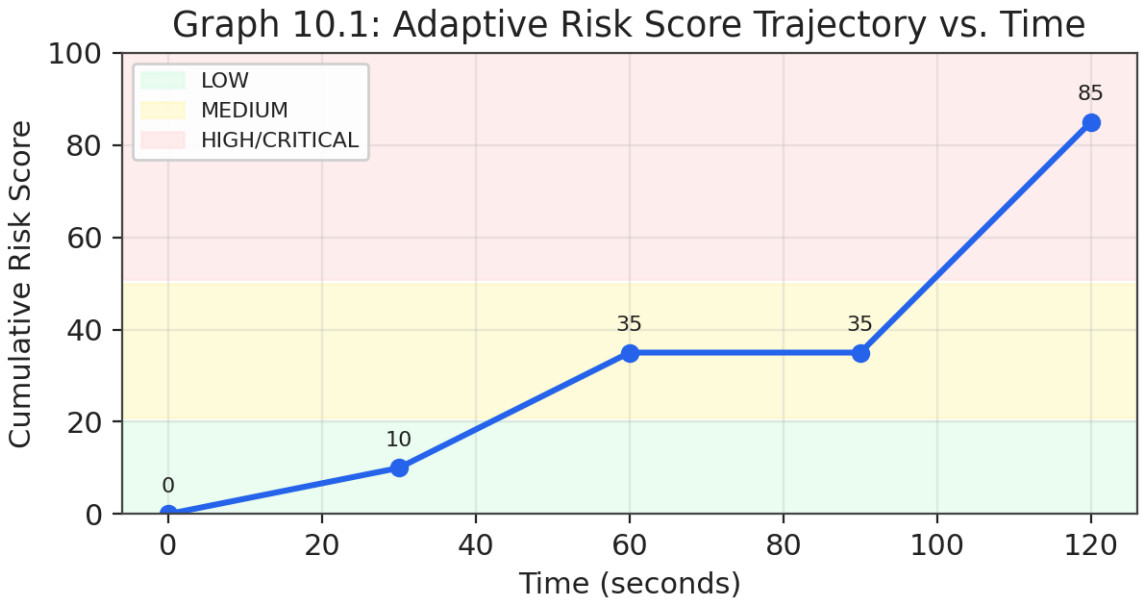


Figure 9.3: Educator Monitoring Dashboard and Live Risk Matrix
Shows candidate cards, real-time risk scores, the proctoring events timeline, and remote session controls.

Figure 9.3
docs/screenshots/02 teacher dashboard.png

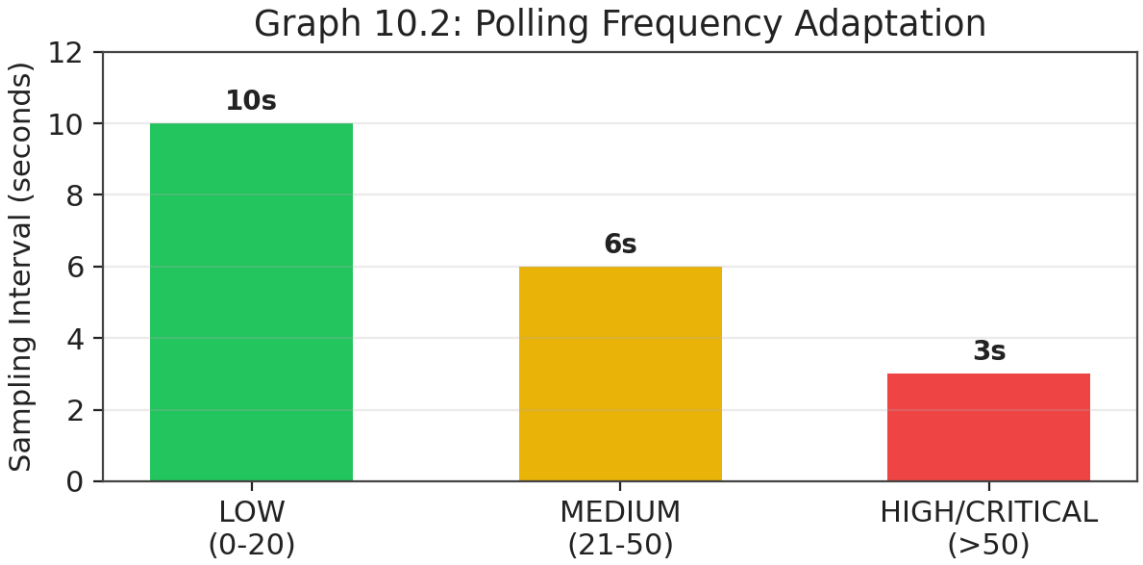


10. Analytical Graphs and Evaluation Metrics



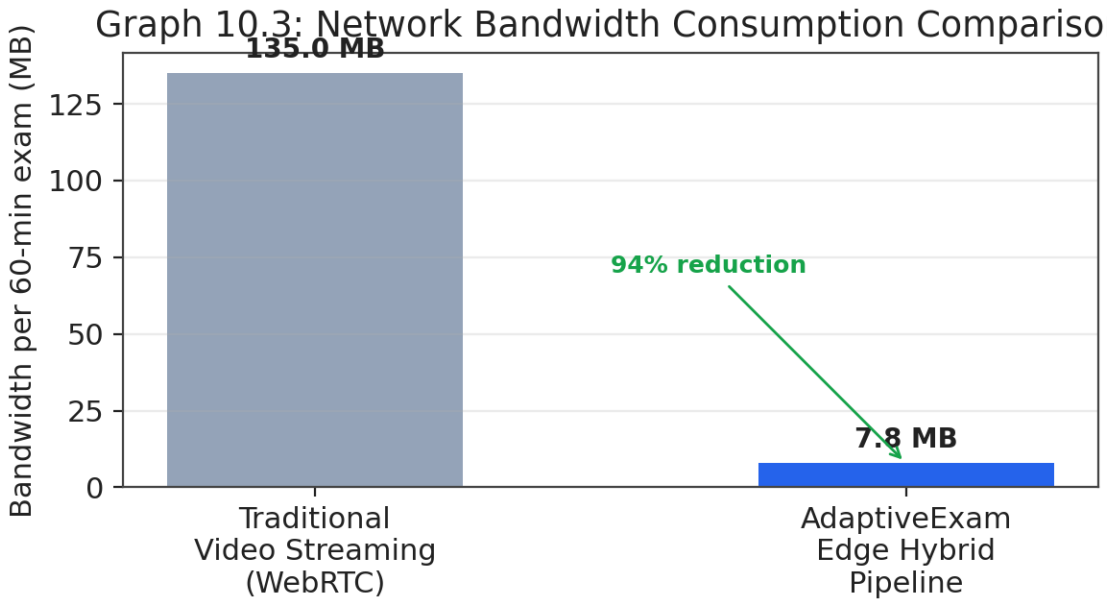
Graph 10.1 — Adaptive Risk Score Trajectory vs. Time

Routine conduct stays at 0; repeated frames of the same phone are deduplicated; subsequent multi-face detection lifts the score into the CRITICAL tier and triggers an SNS alert.



Graph 10.2 — Polling Frequency Adaptation

AdaptiveExam keeps a relaxed 10-second cadence during normal conduct and accelerates to 3 seconds only when suspicious behavior is observed.



Graph 10.3 — Network Bandwidth Consumption Comparison

Running TensorFlow.js locally and transmitting only periodic JSON telemetry and flagged evidence images cuts bandwidth from ~135 MB to under 8 MB per student-hour.

11. Conclusion

AdaptiveExam bridges the gap between scalable remote evaluation and rigorous academic integrity by combining browser-native edge AI inference with AWS serverless services. It overcomes the main weaknesses of existing proctoring architectures:

1. No cloud video bottleneck: COCO-SSD at the edge gives instant prohibited-device recognition while cutting bandwidth by over 94%.
2. Resilient dual-tier architecture: when cloud APIs hit permission or rate limits, local proctoring continues without failing.
3. Equitable, intelligent scoring: grace-period absence filtering, stateful deduplication, and dynamic polling prevent false-positive disqualifications.
4. Economical scalability: built on Cognito, S3, Rekognition, Lambda, DynamoDB, and SNS, the platform has zero idle server cost and scales automatically during peak exam windows.
5. Actionable educator experience: instructors get live forensic telemetry, photographic evidence in S3, and real-time SNS email notifications for decisive intervention.

12. Future Work

1. 3D head pose and eye-gaze estimation (MediaPipe FaceMesh): 468-point landmark tracking in the browser to measure yaw, pitch, and roll and detect subtle off-screen reading without extra hardware.
2. Secondary smartphone camera via WebRTC: the candidate scans a QR code to pair a phone as a 45-degree side-angle camera for full desk and keyboard visibility.
3. Keystroke dynamics and typing biometrics: analyze dwell and flight times during essay answers to detect cut-and-paste behavior or secondary typists.
4. Offline-first synchronization (PWA with IndexedDB): cache progress and queue encrypted telemetry locally so a temporary Wi-Fi loss does not interrupt the exam, syncing on reconnection.

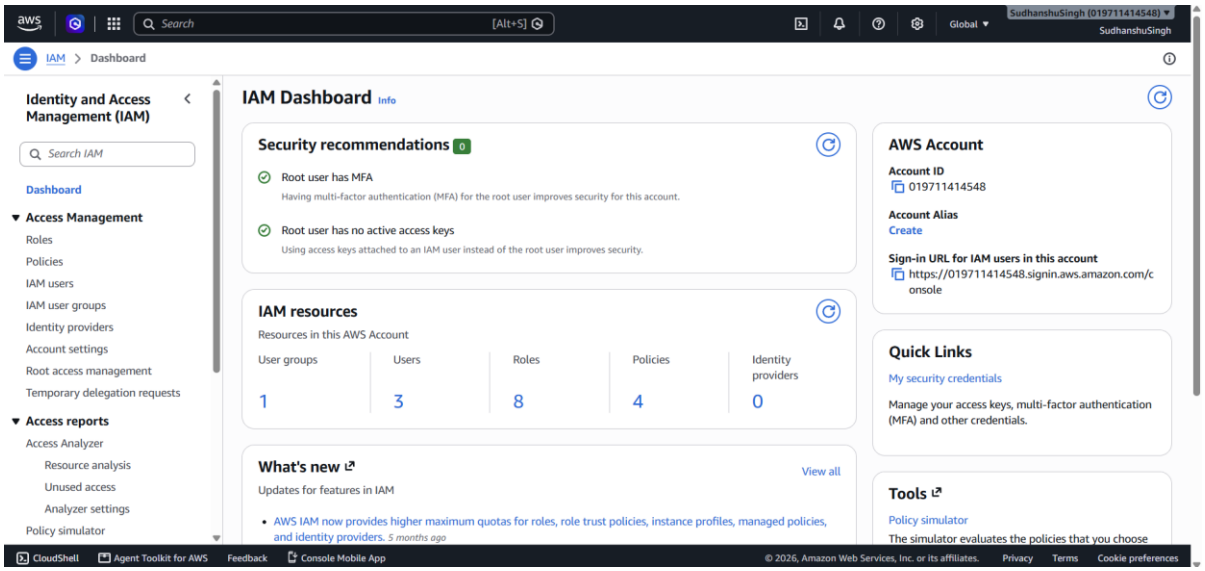
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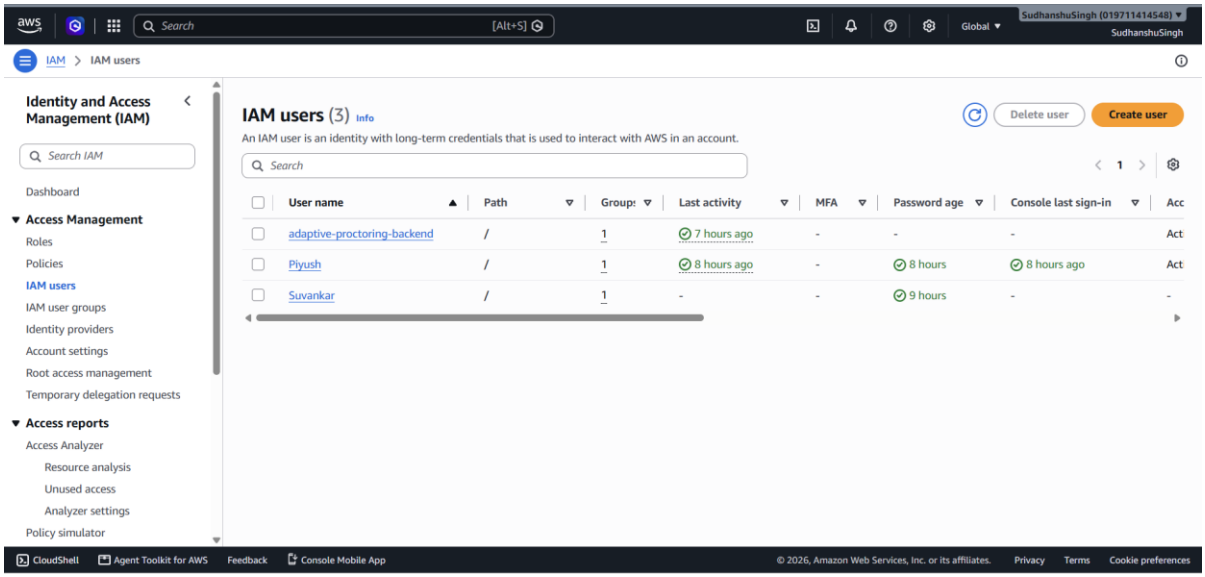
14. Appendix A: AWS Service Configuration and Application Screenshots

This appendix documents the deployed AWS console configuration for each managed service and the complete end-to-end teacher and student application workflow, captured directly from the live AdaptiveExam deployment.

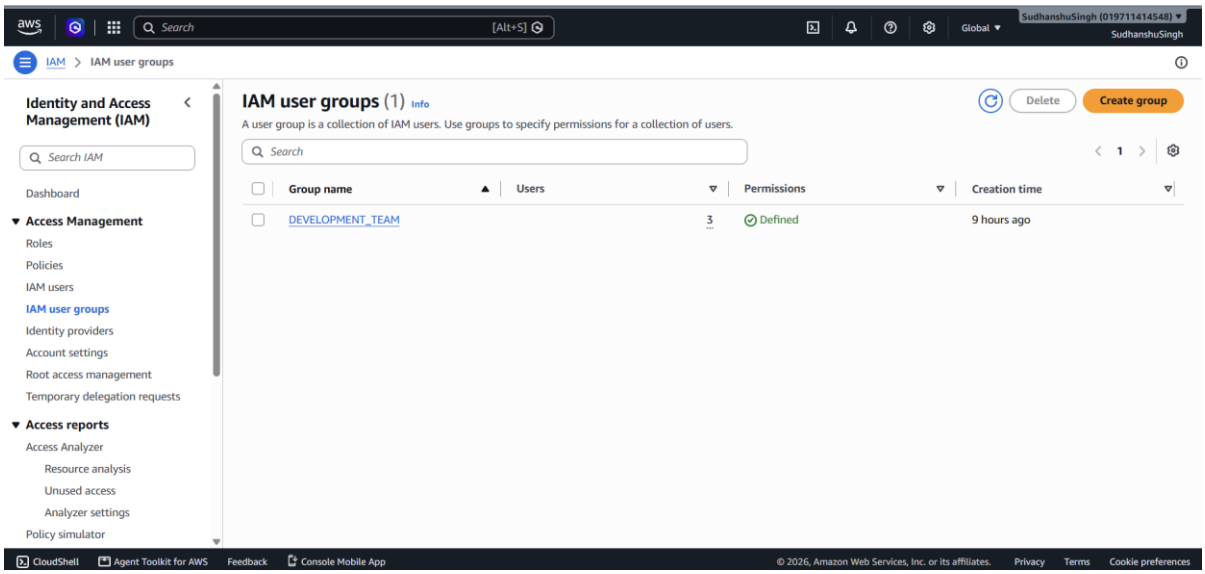
A.1 Amazon Cognito IAM and Identity Pool Configuration



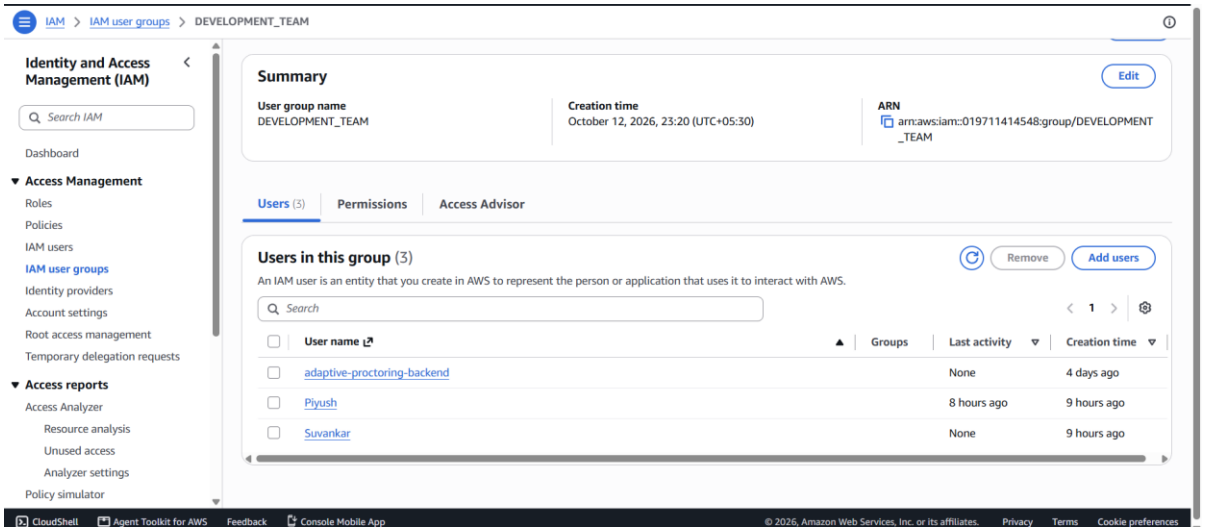
A.1.1 — IAM / Cognito identity configuration



A.1.2 — IAM / Cognito identity configuration

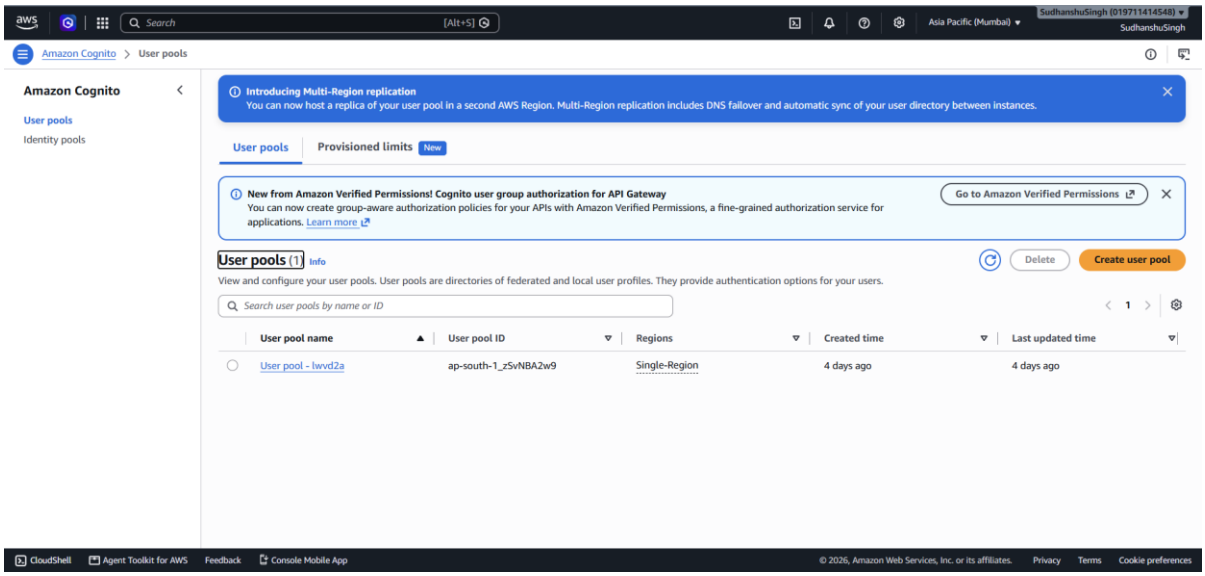


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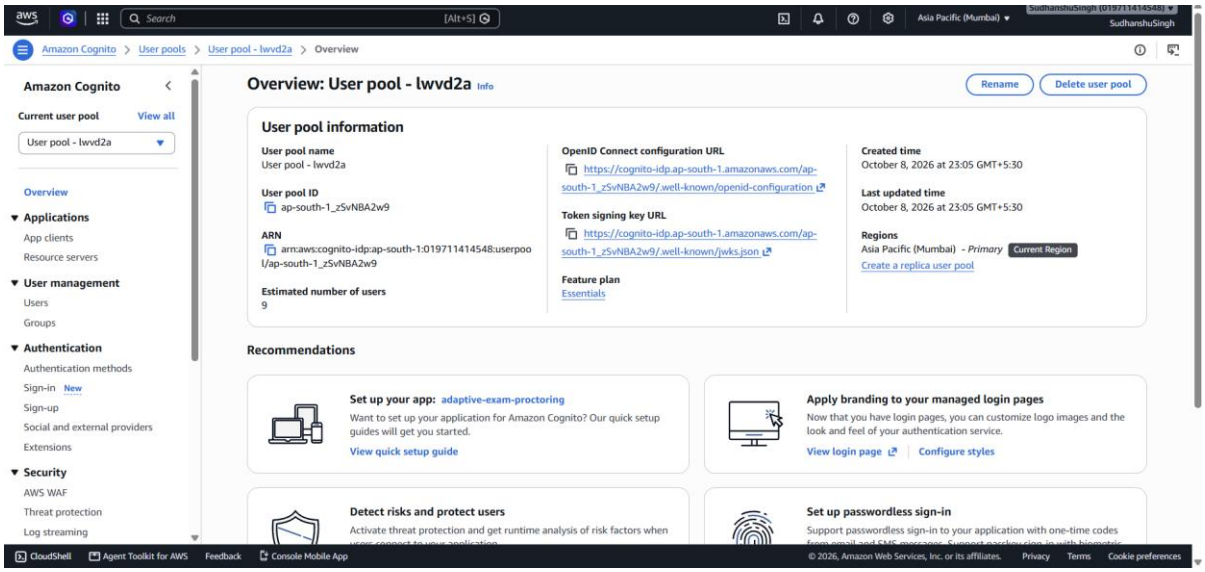


A.1.4 — IAM / Cognito identity configuration

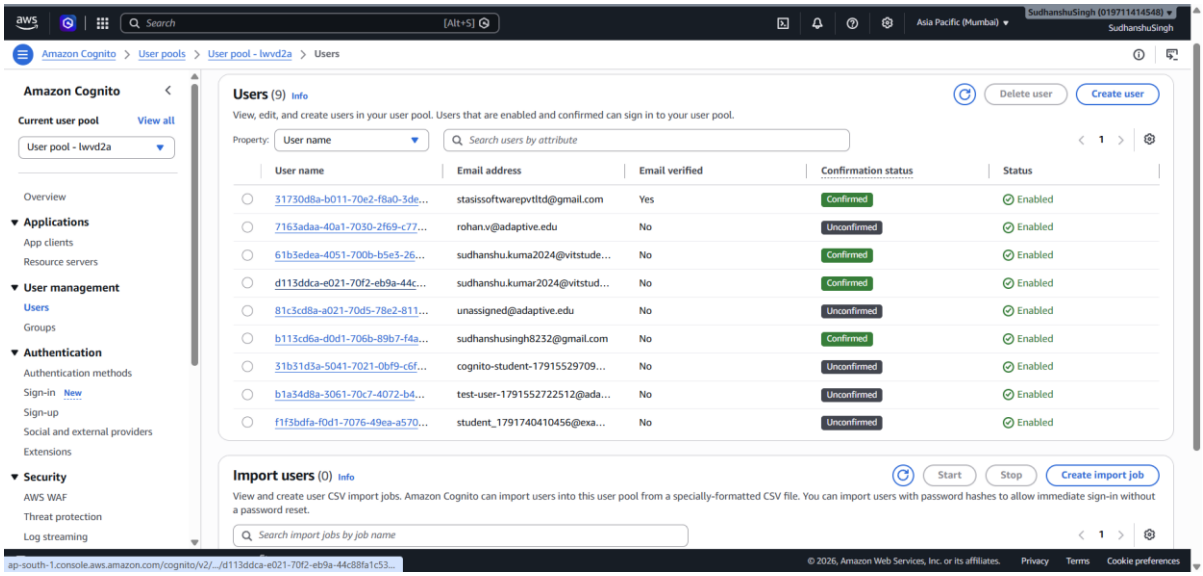
A.2 Amazon Cognito User Pool Configuration



A.2.1 — Cognito user pool console

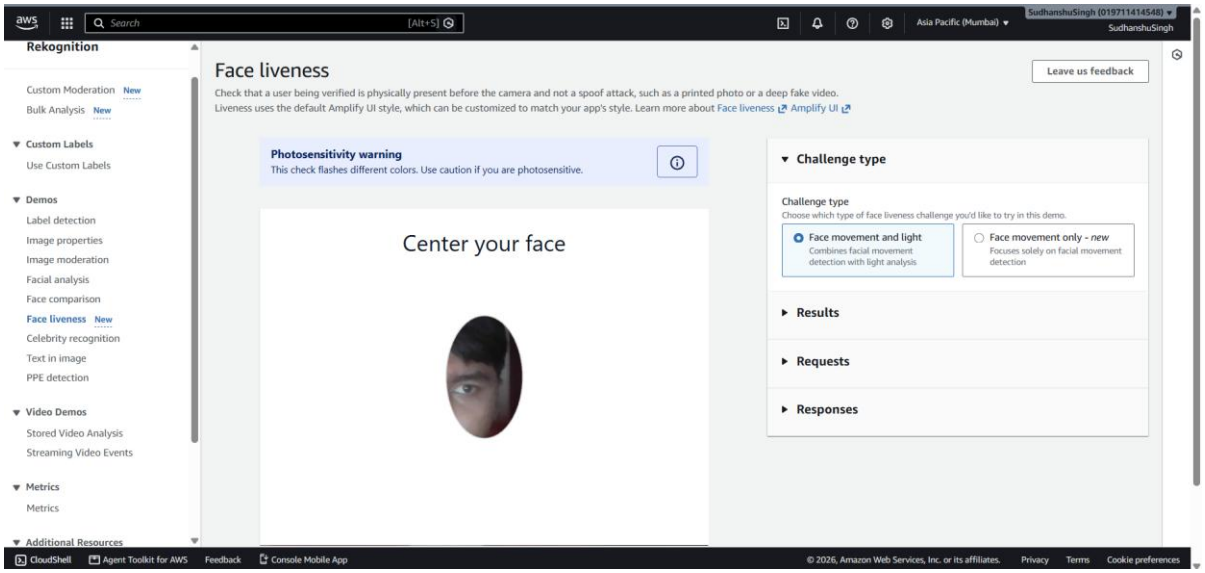


A.2.2 — Cognito user pool console

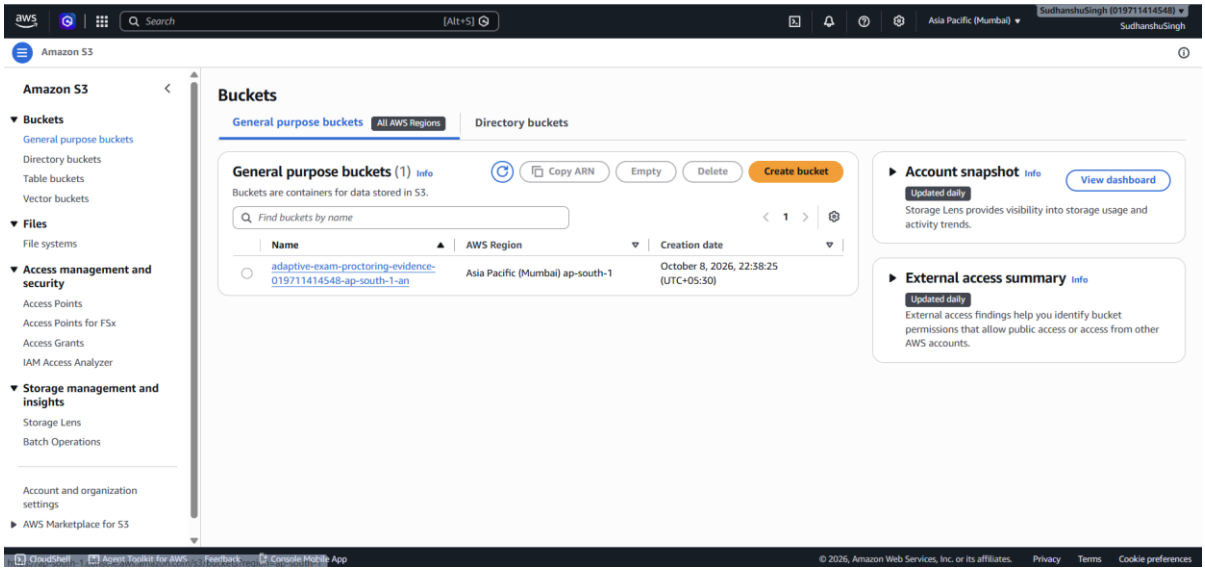


A.2.3 — Cognito user pool console

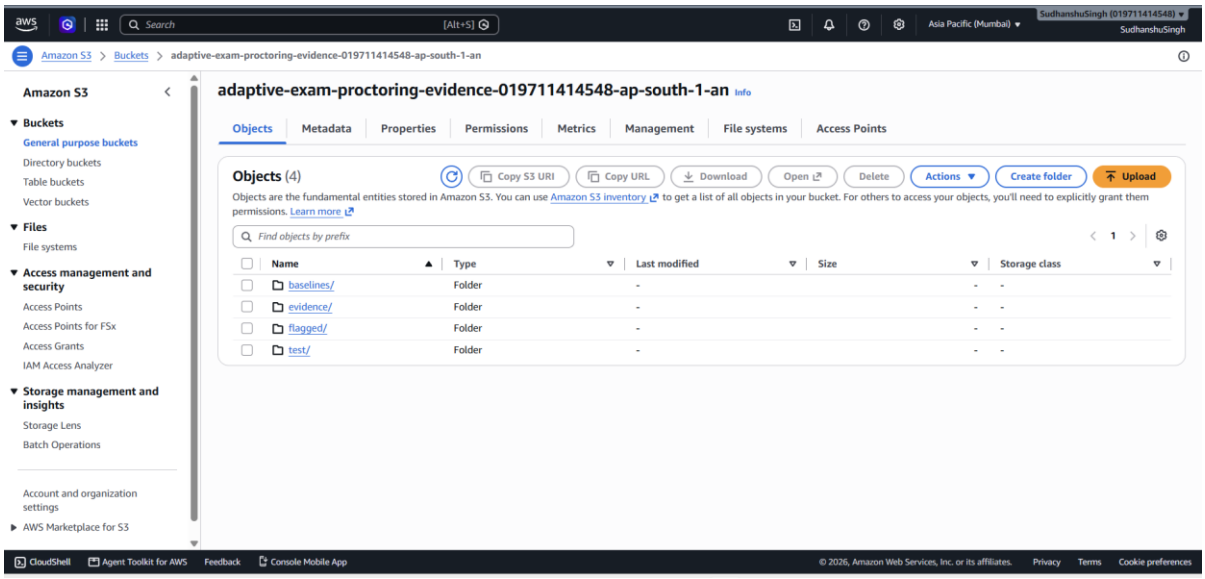
A.3 Amazon Rekognition Configuration



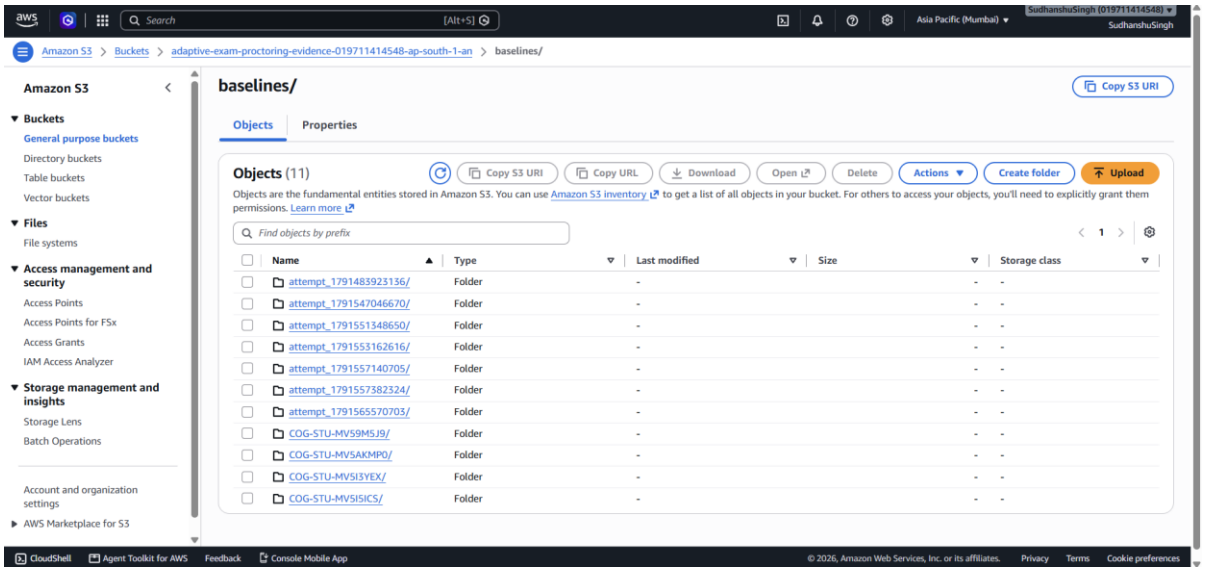
A.3.1 — Rekognition face collection console



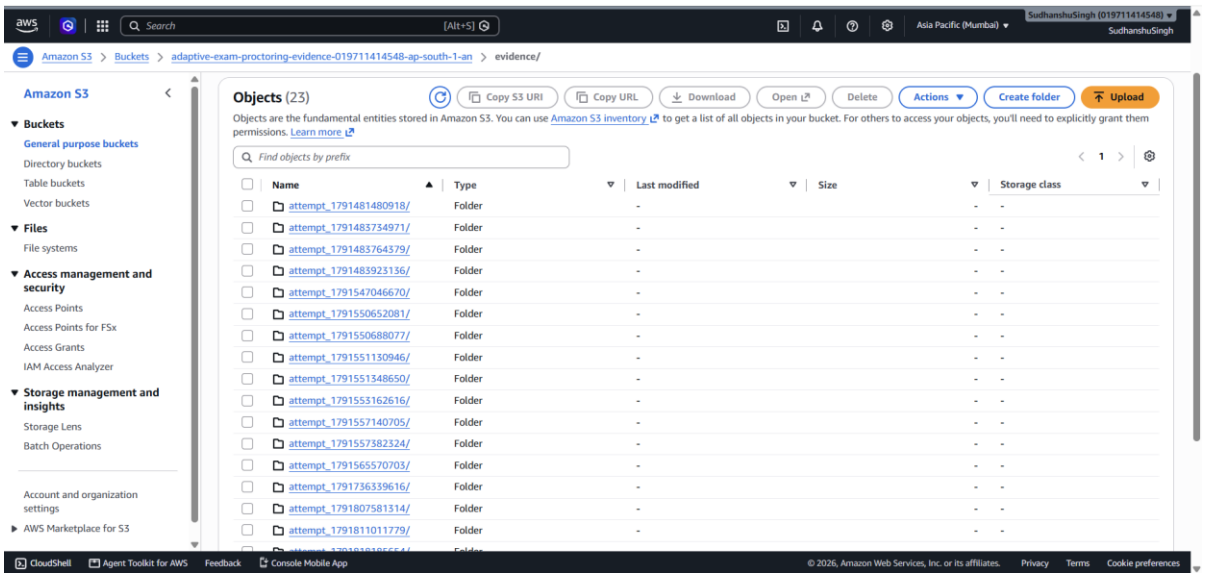
A.3.2 — Rekognition face collection console



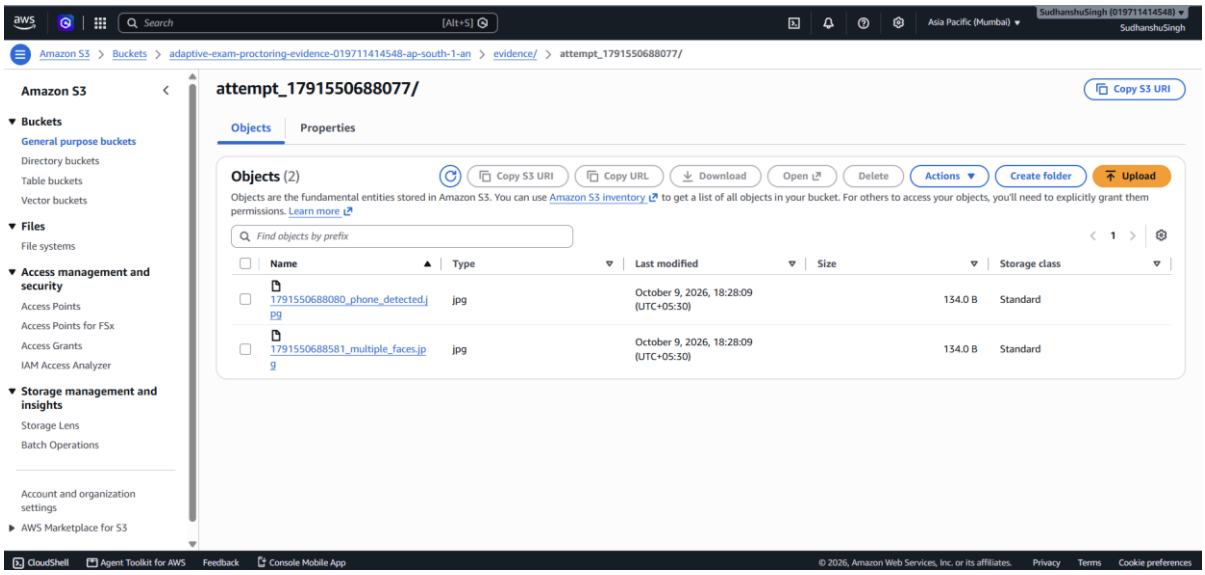
A.3.3 — Rekognition face collection console



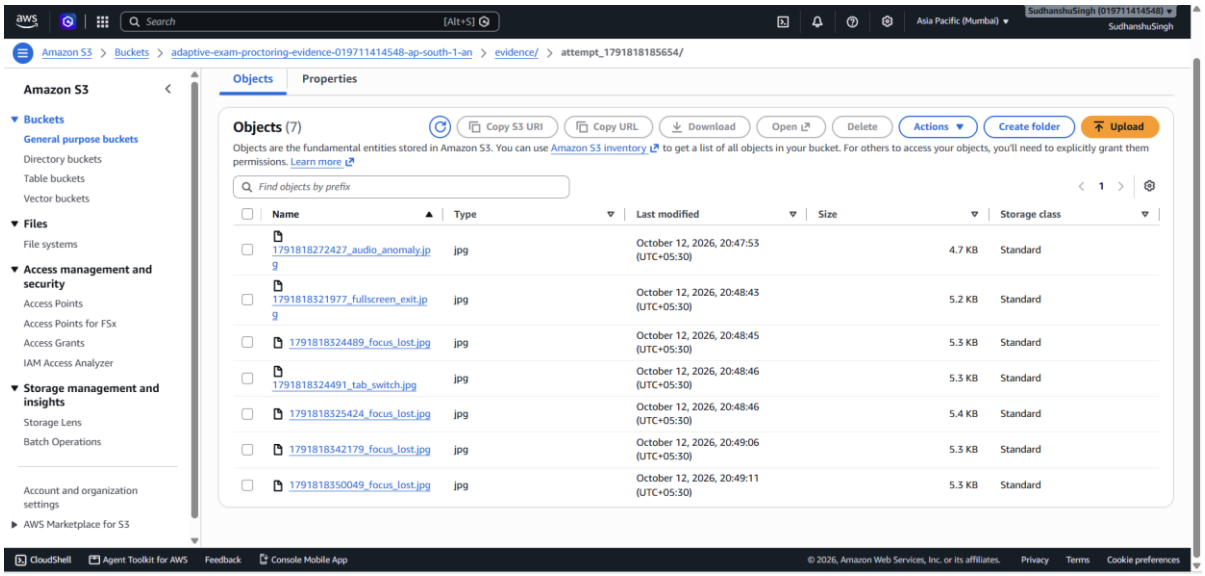
A.3.4 — Rekognition face collection console



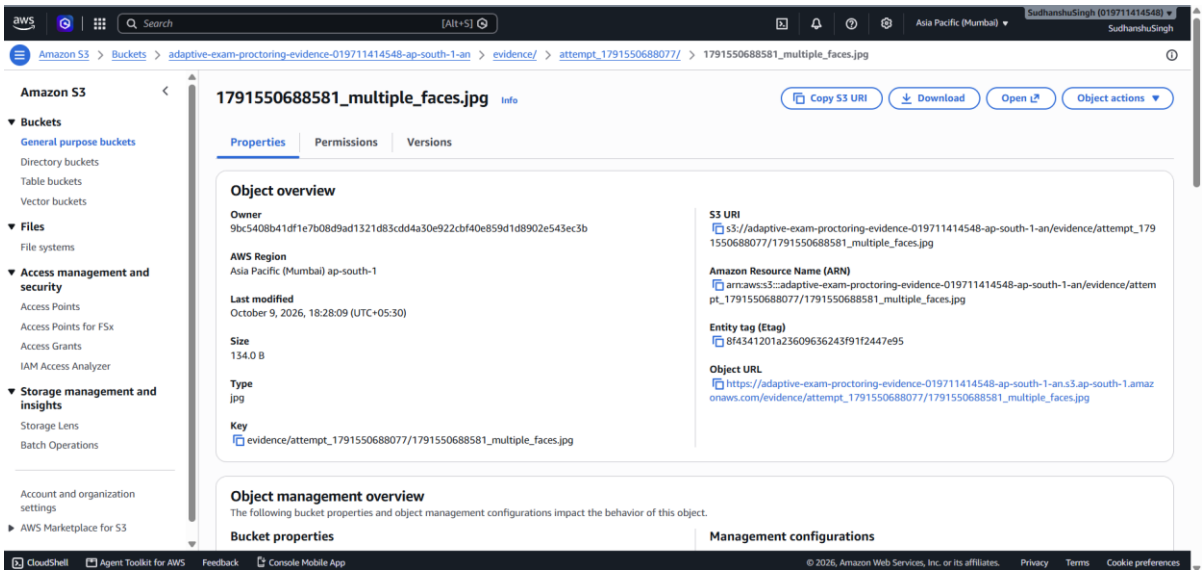
A.3.5 — Rekognition face collection console



A.3.6 — Rekognition face collection console

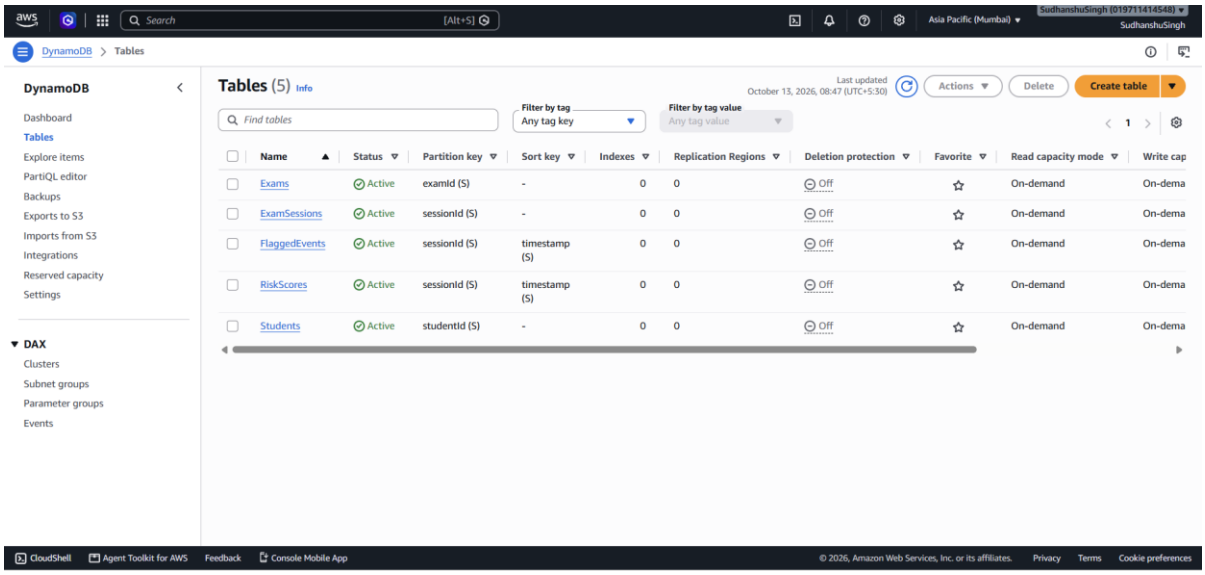


A.3.7 — Rekognition face collection console

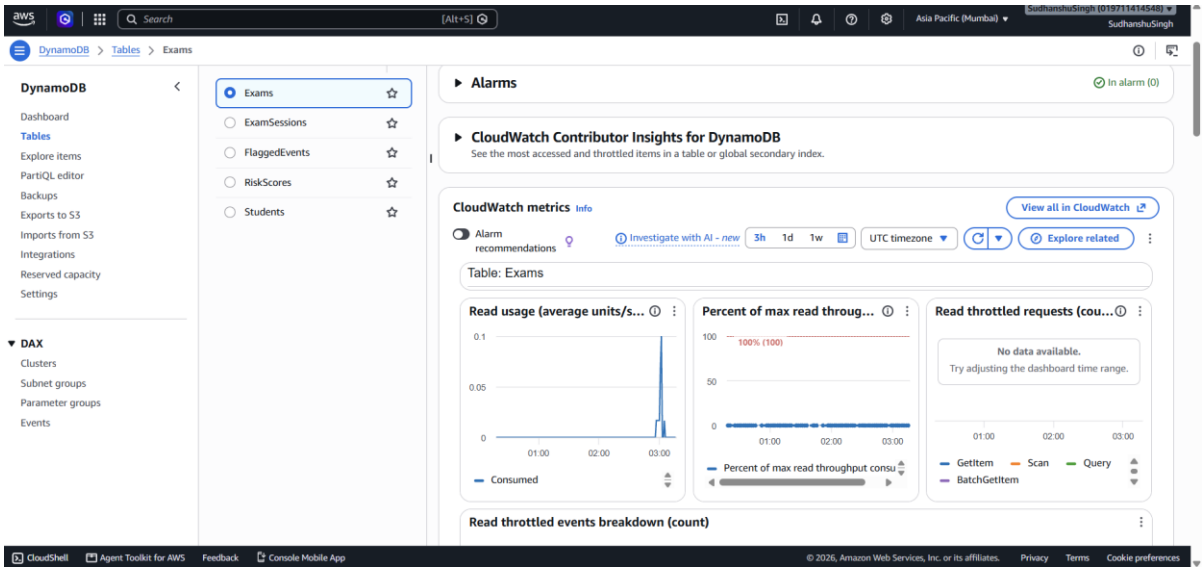


A.3.8 — detected multiple faces – image evidence

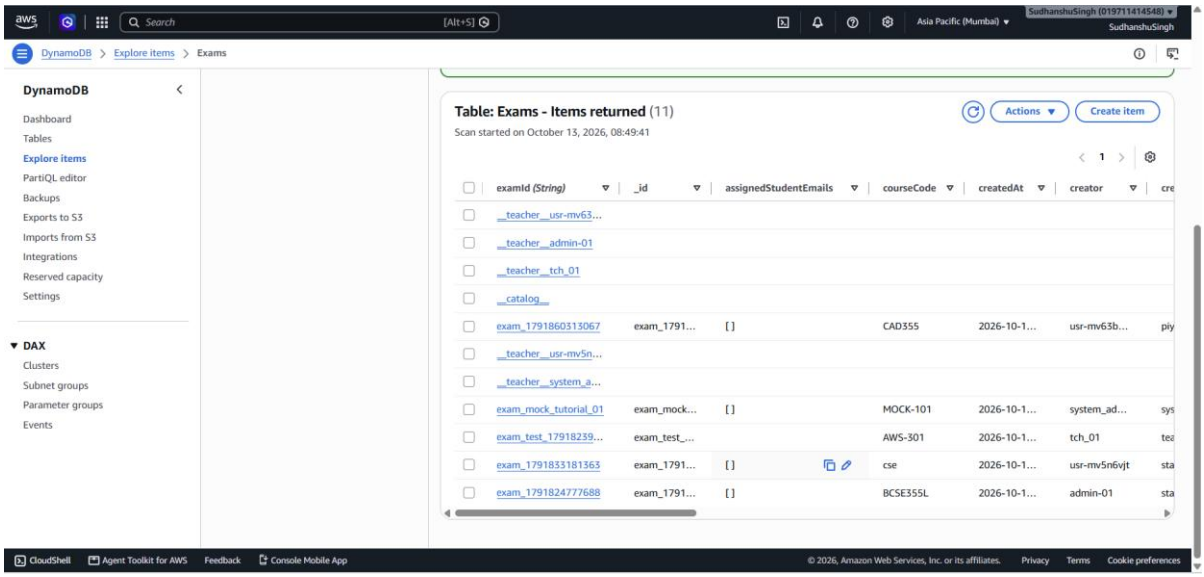
A.4 DynamoDB Database Configuration



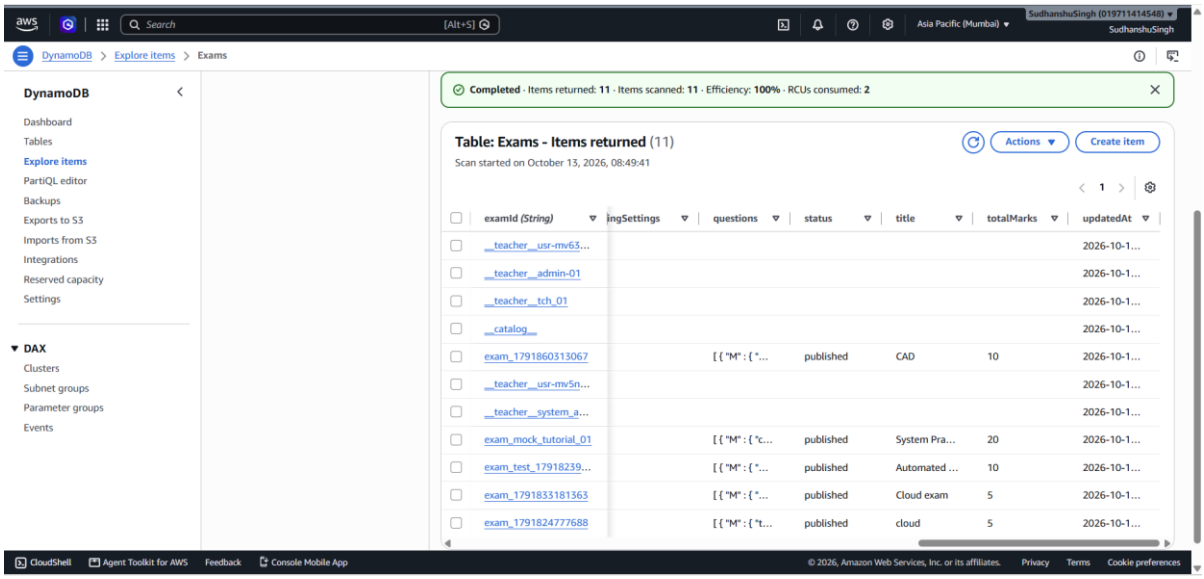
A.4.2 —database console



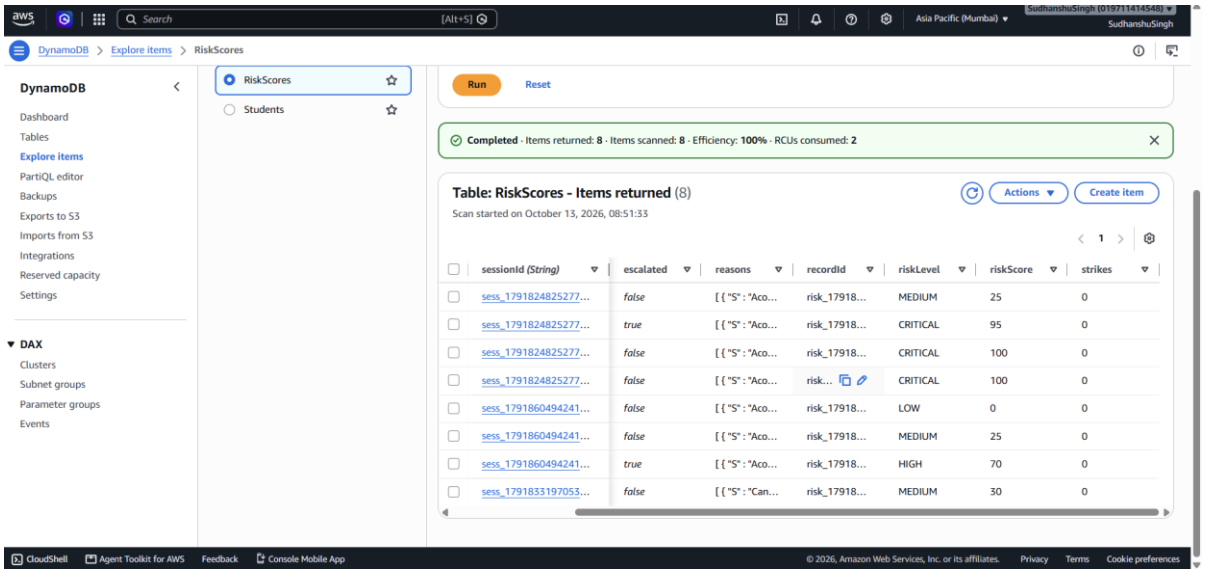
A.4.3 — database console



A.4.4 — database console

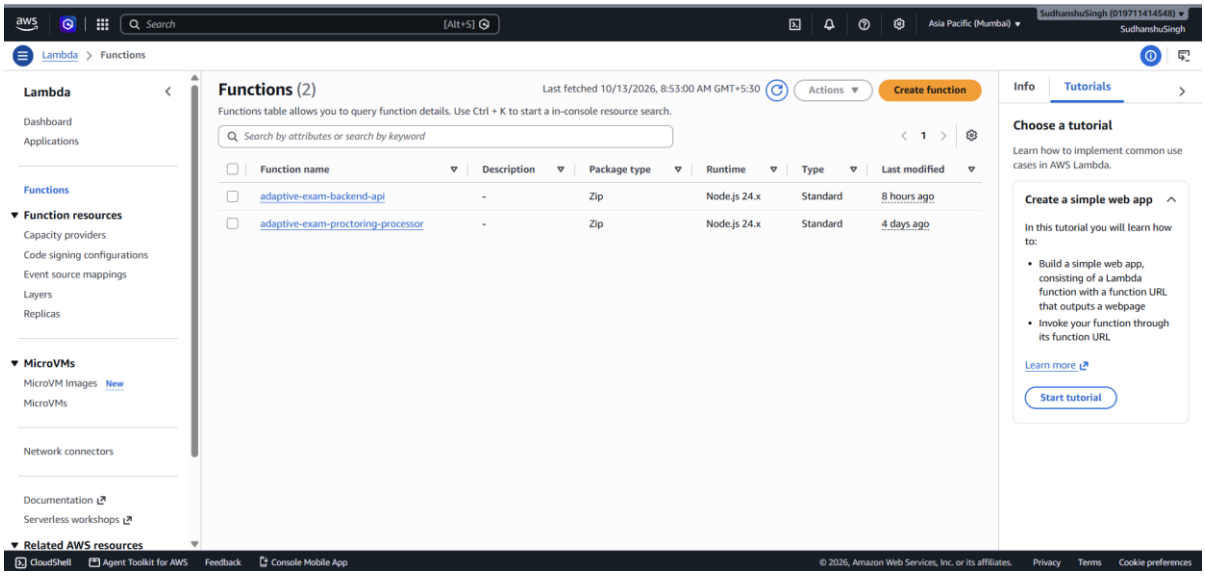


A.4.5 — database console

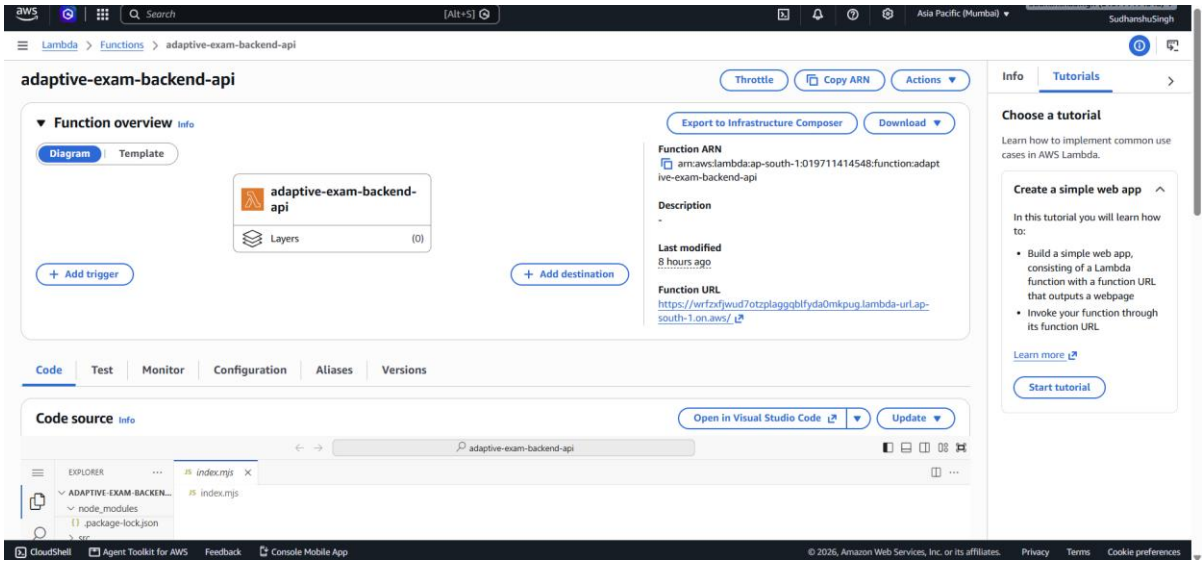


A.4.6 — database console

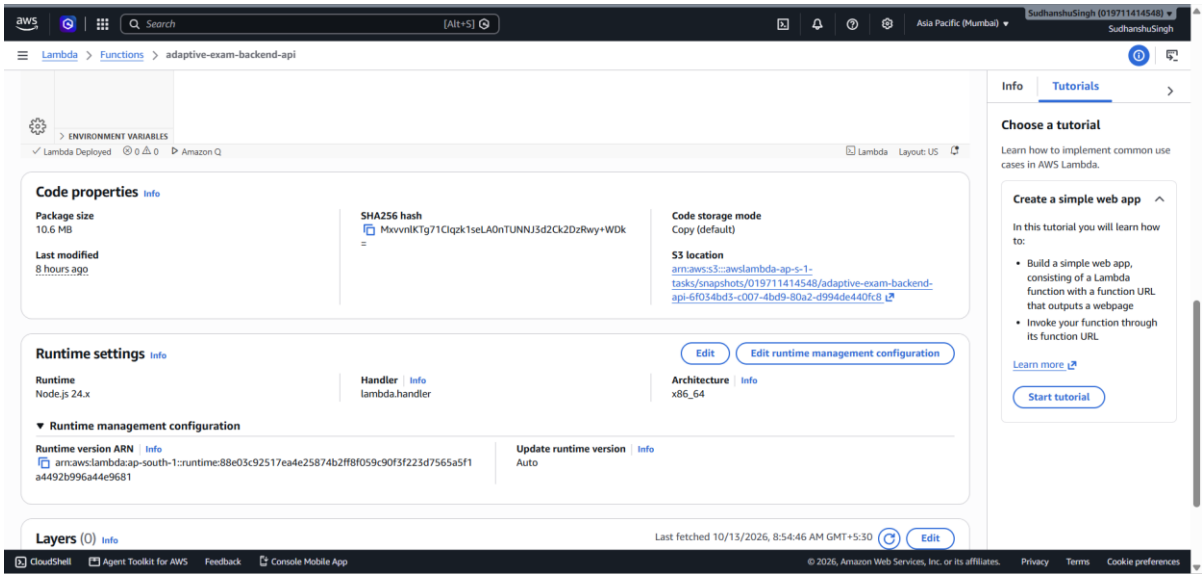
A.5 AWS Lambda Function Configuration



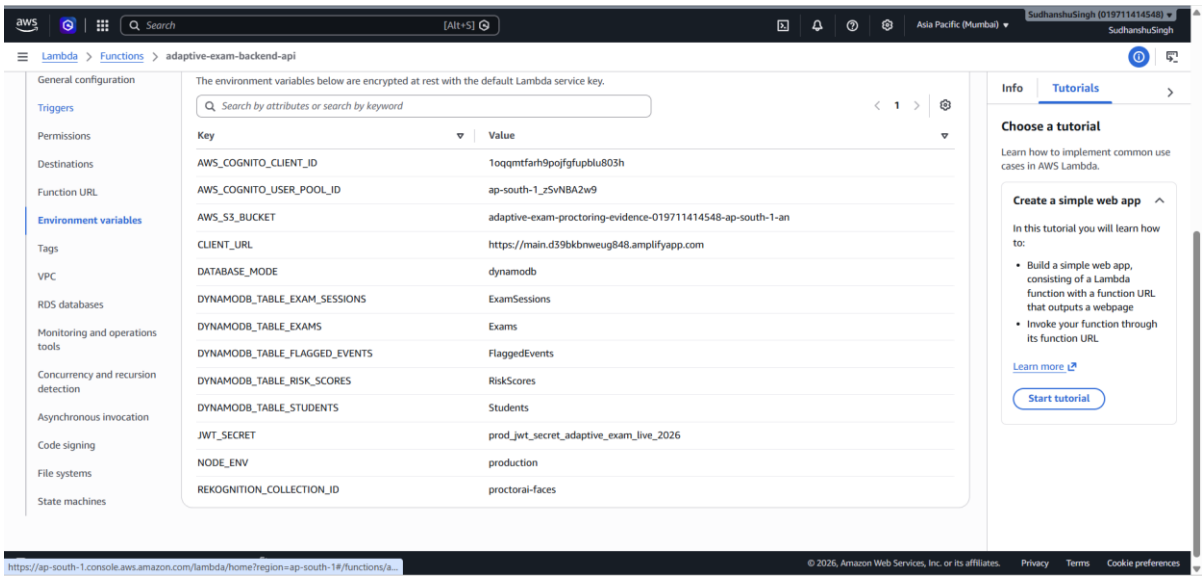
A.5.1 — Lambda handlers console



A.5.2— Lambda handlers console



A.5.3 — Lambda handlers console

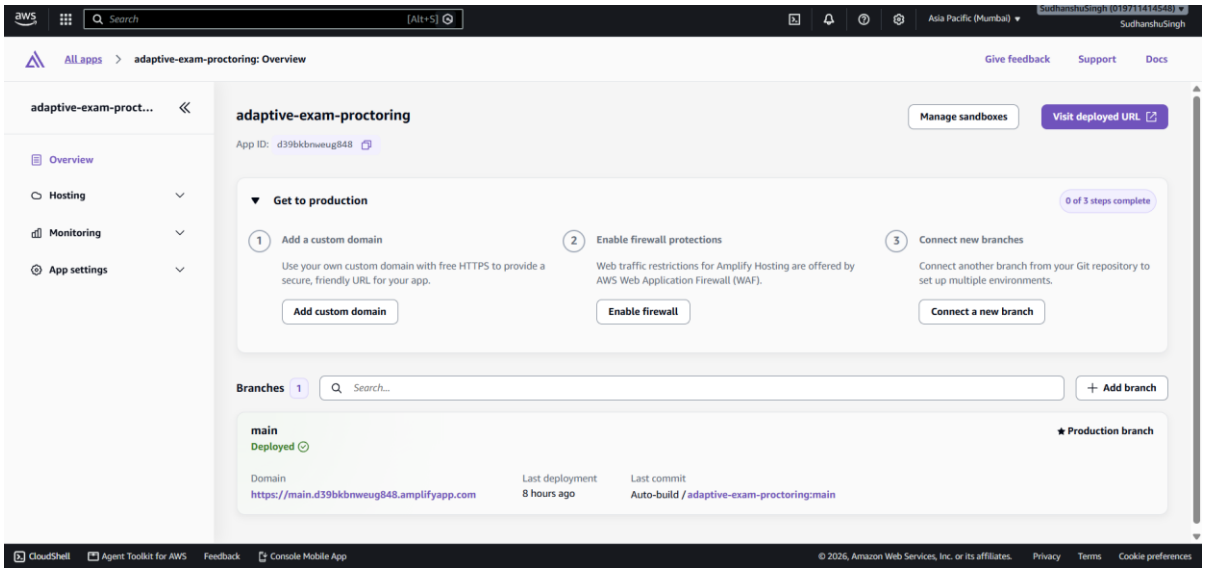


A.5.4 — Lambda handlers console

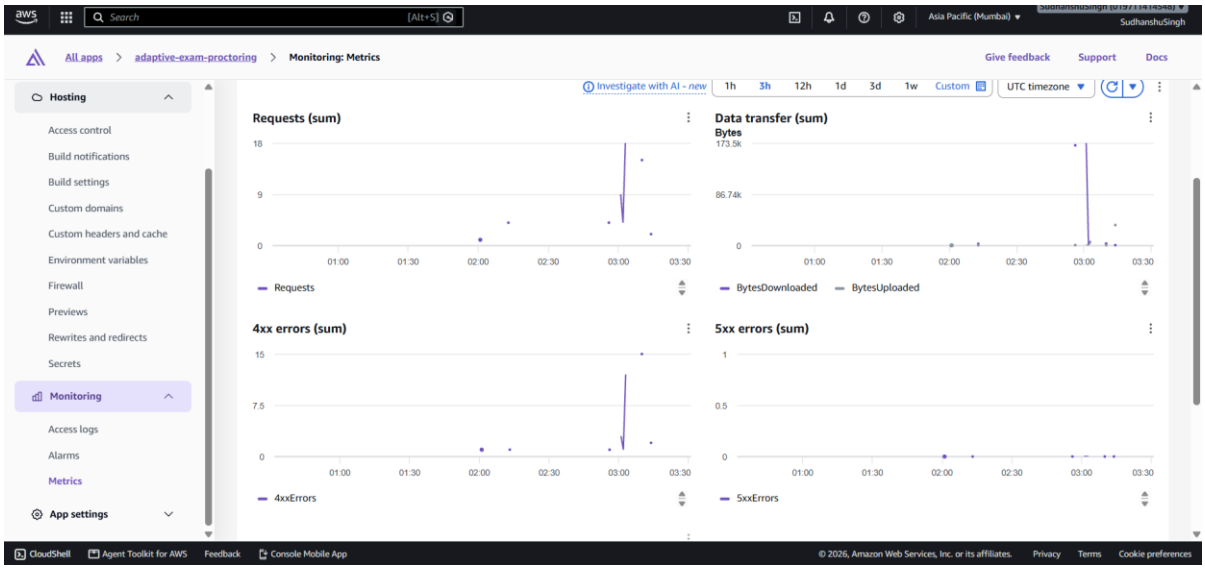


A.5.5— Lambda handlers console

A.6 AWS Amplify Deployment

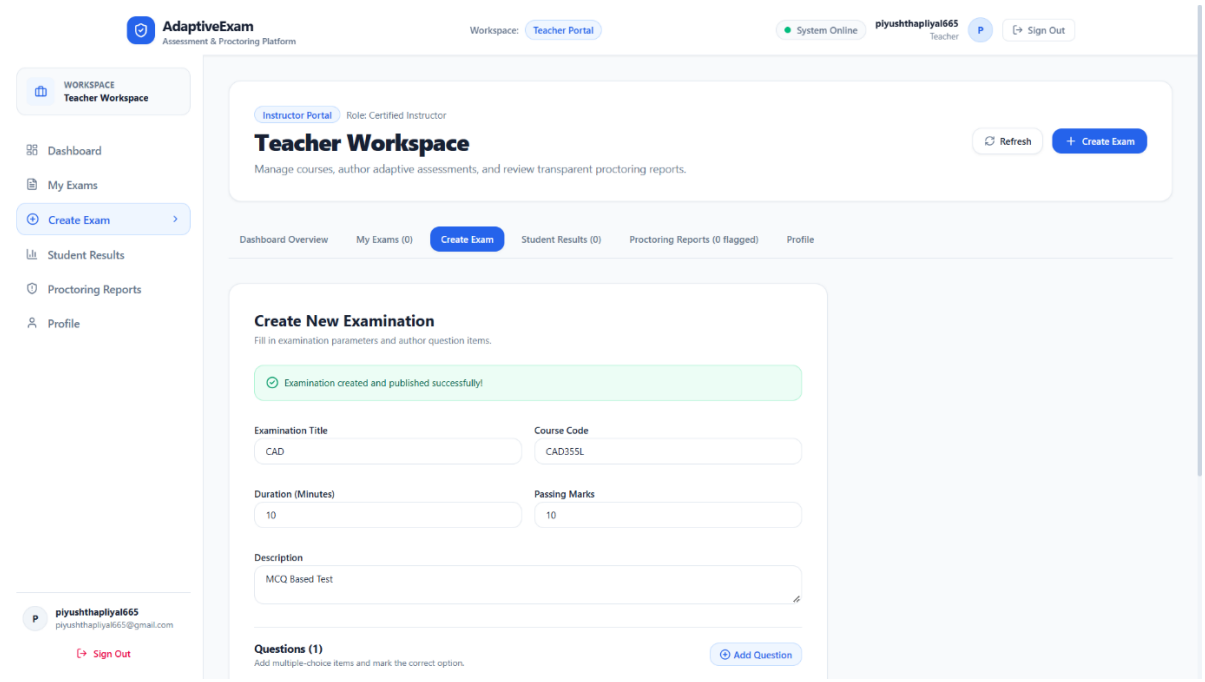


A.6.1 — Amplify hosting console

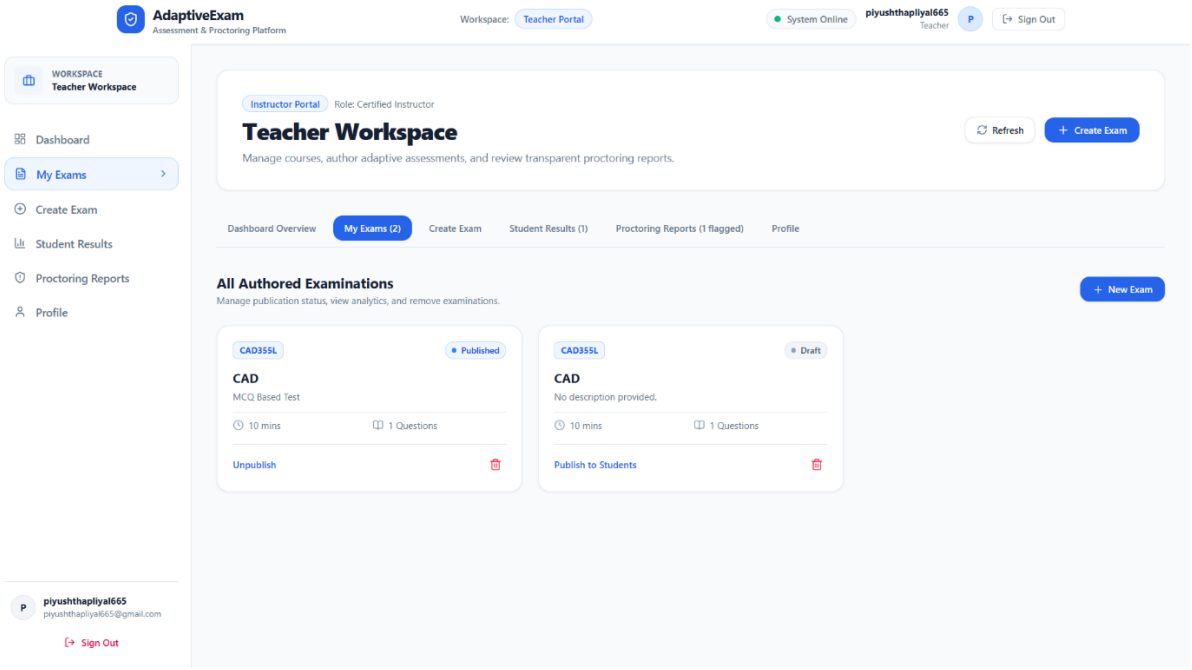


A.6.2 — Amplify hosting console

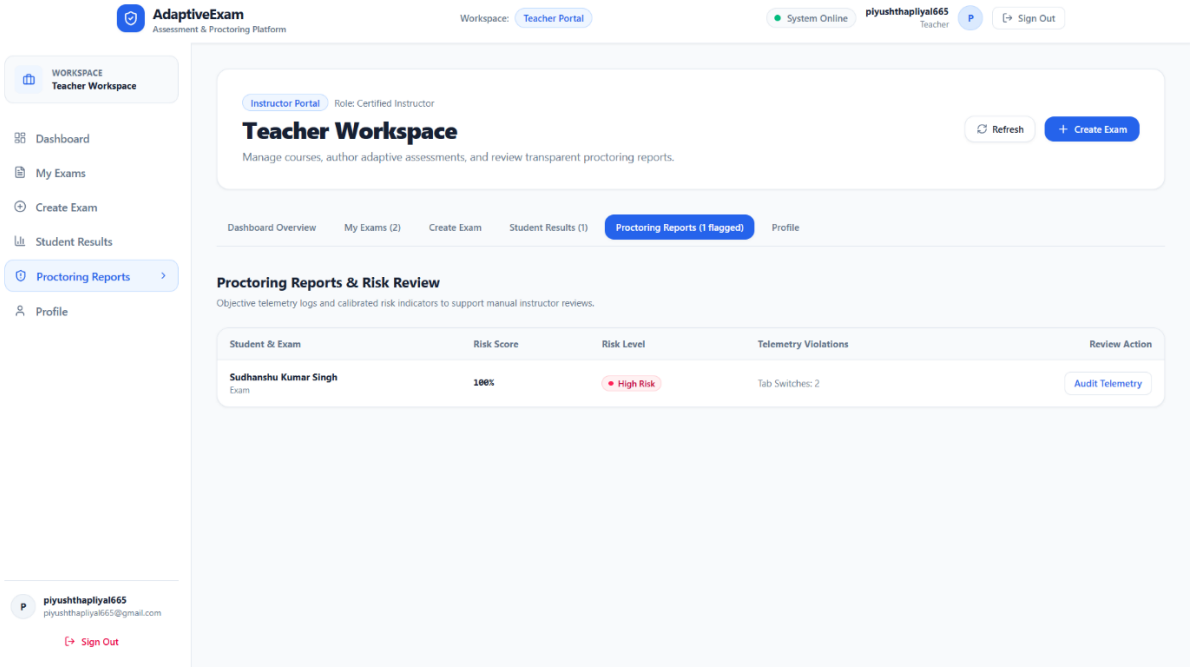
A.7 Teacher Portal Walkthrough



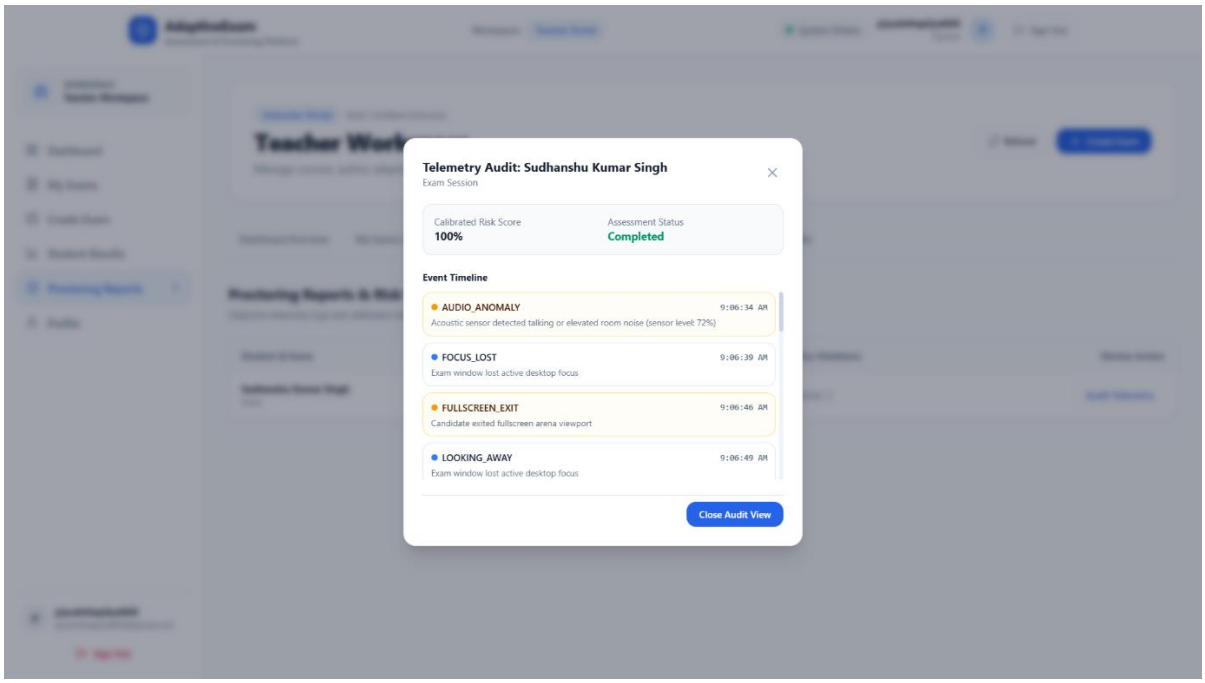
A.7.1 — Teacher Workspace screens



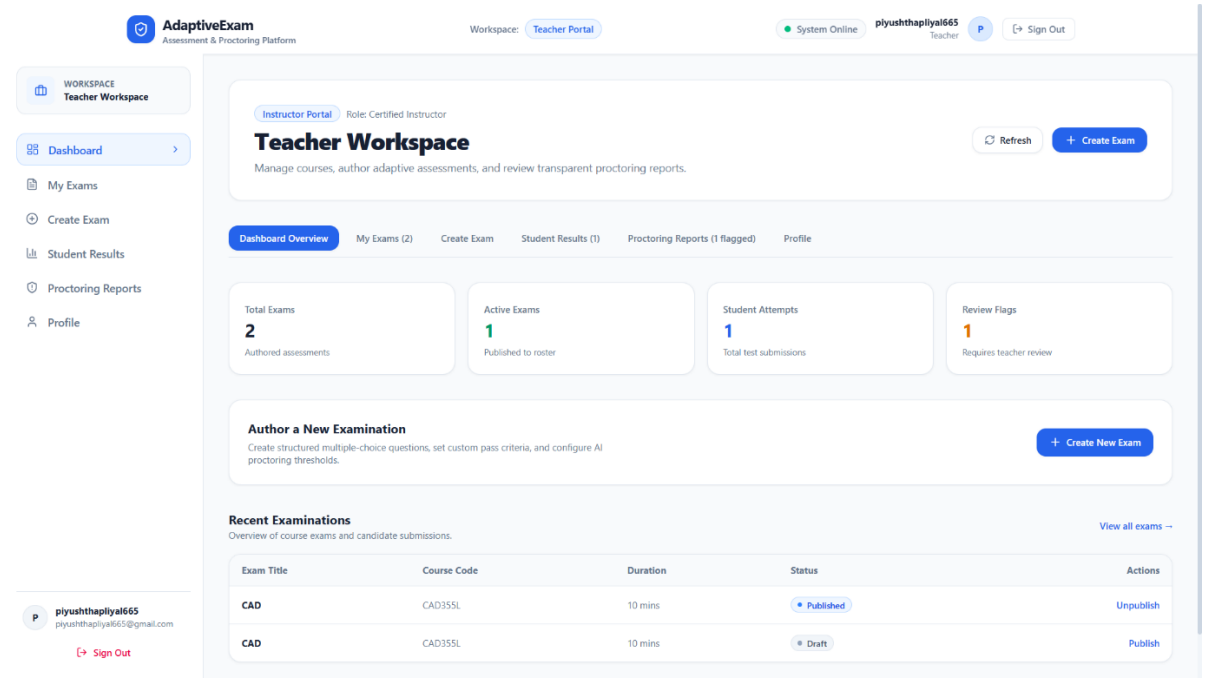
A.7.2 — Teacher Workspace screens



A.7.3 — Teacher Workspace screens

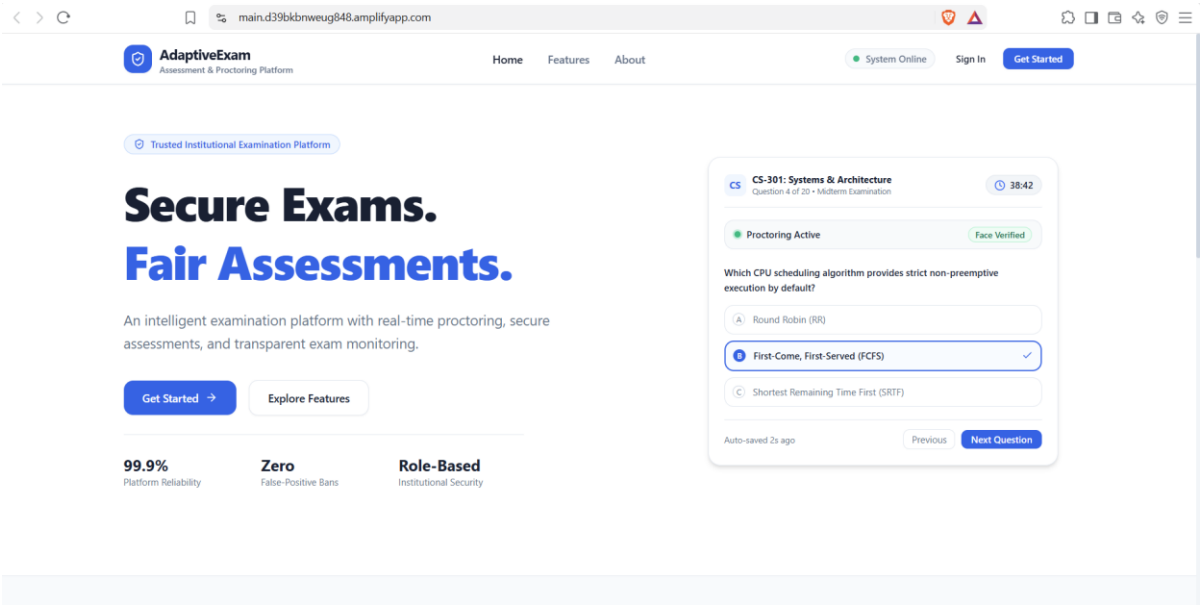


A.7.4 — Teacher Workspace screens

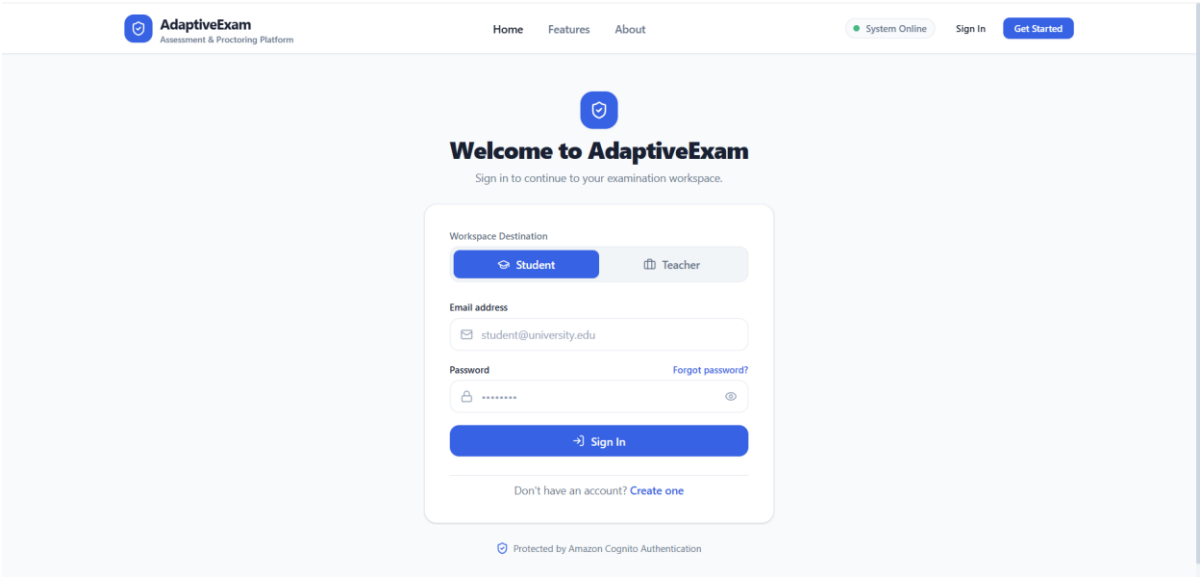


A.7.5 — Teacher Workspace screens

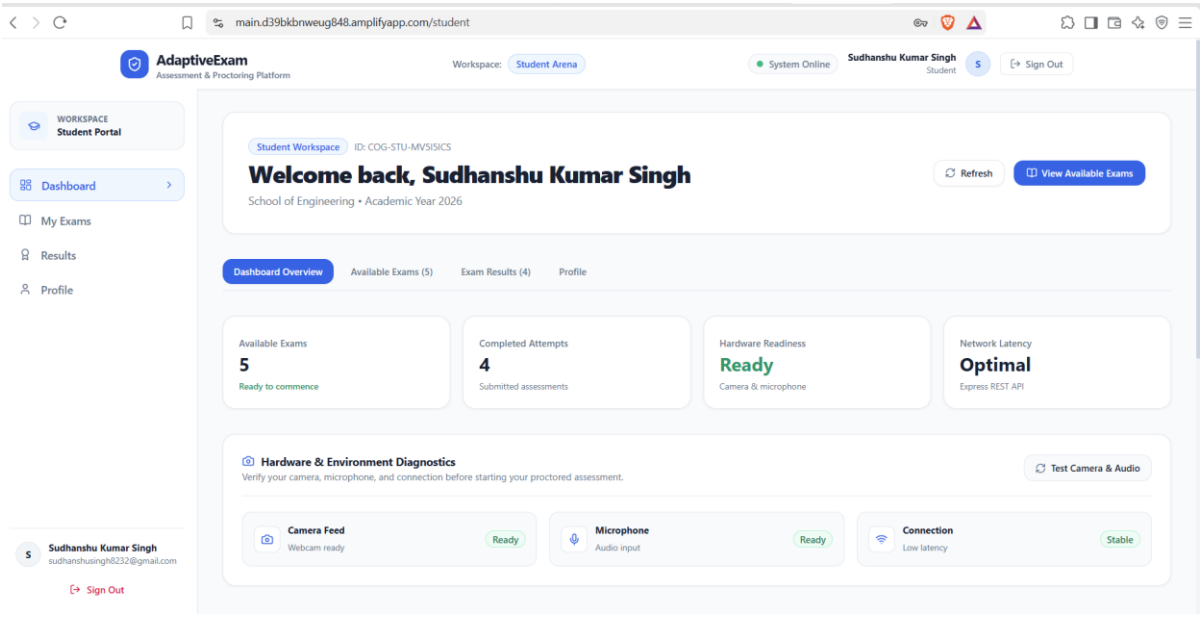
A.8 Student Portal Walkthrough



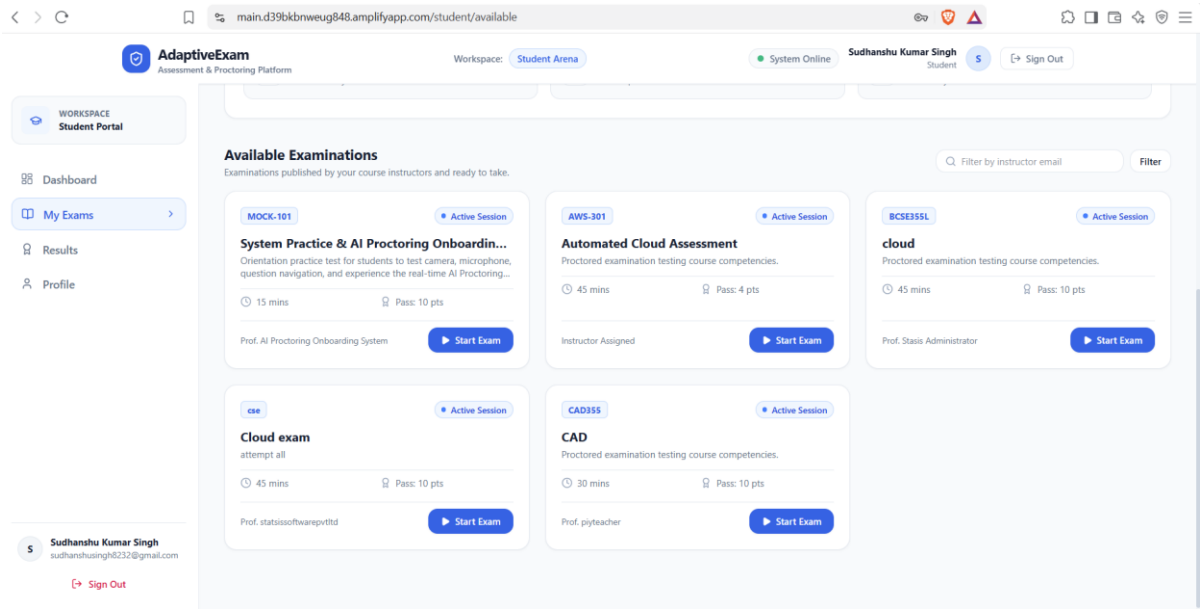
A.8.1 — Student Arena screens



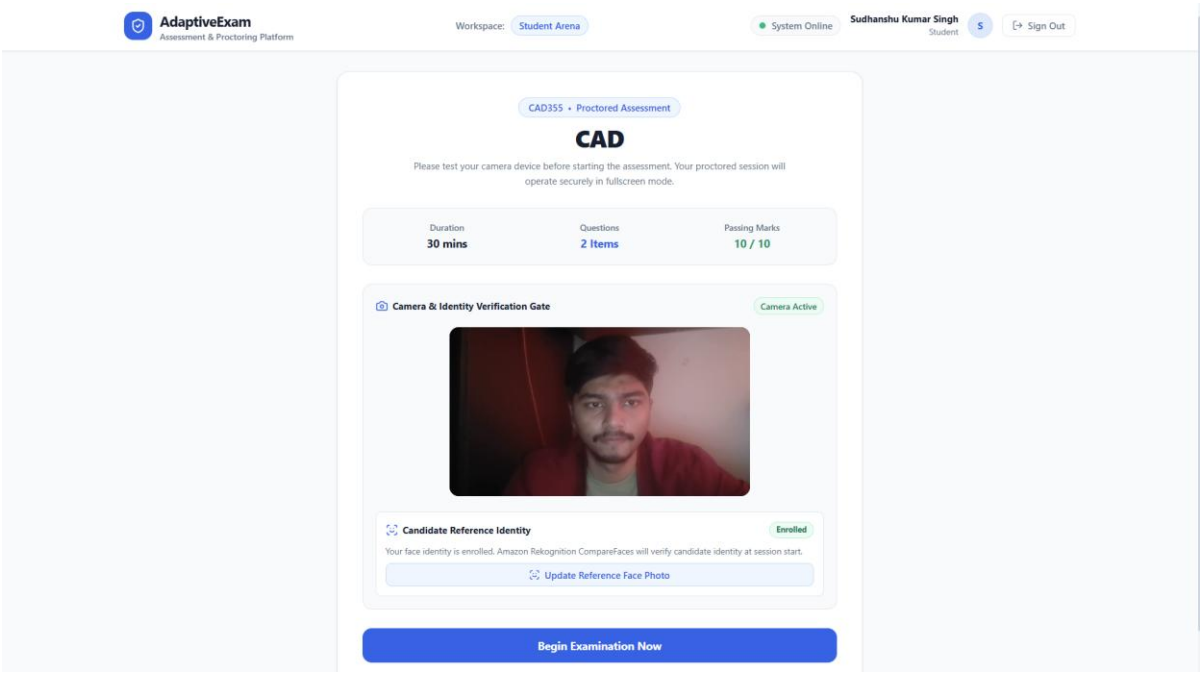
A.8.2 — Student Arena screens



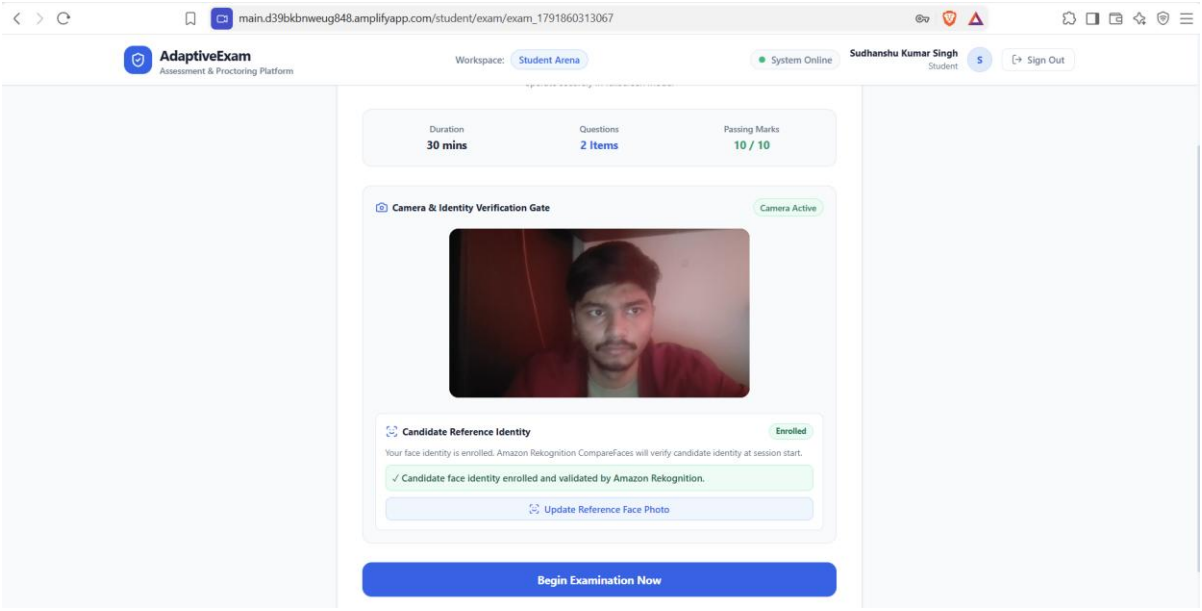
A.8.3 — Student Arena screens



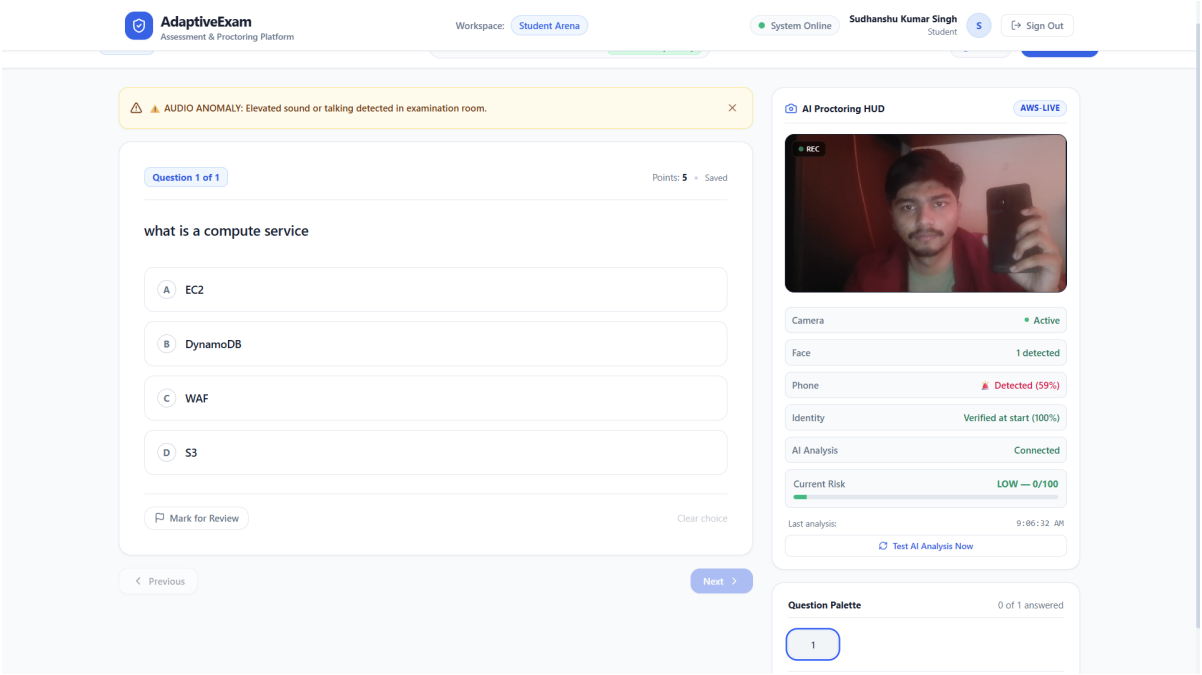
A.8.4 — Student Arena screens



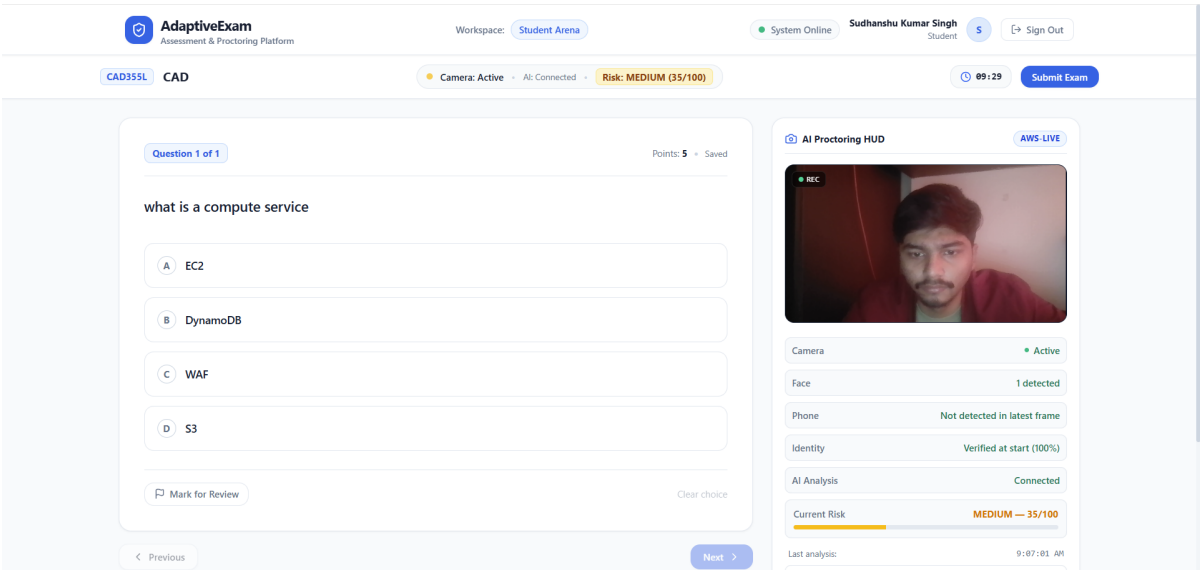
A.8.5 — Student Arena screens



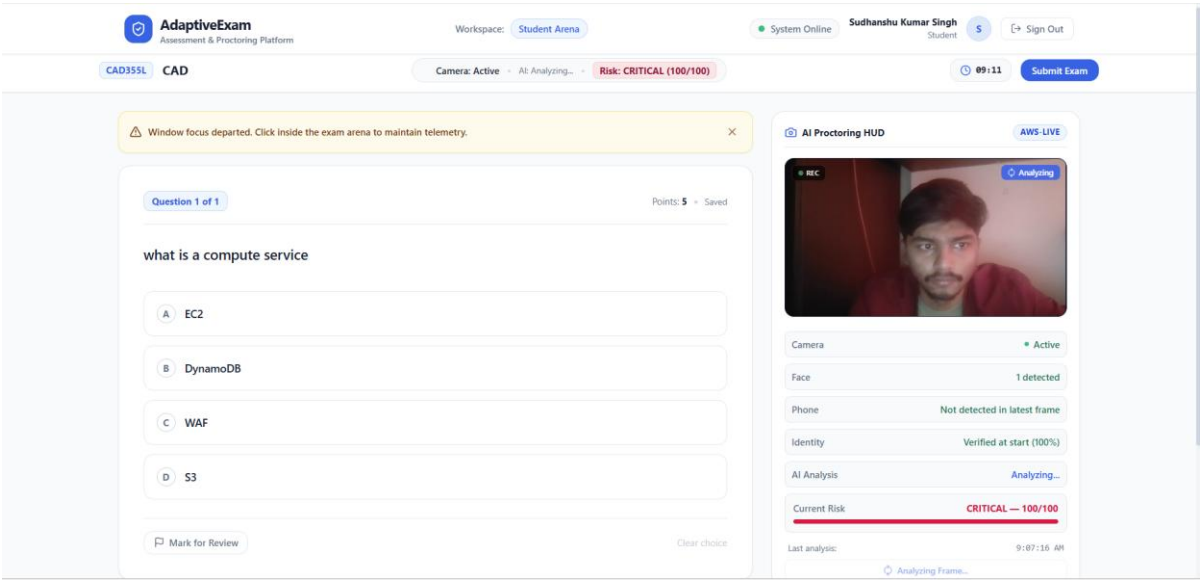
A.8.6 — Student Arena screens



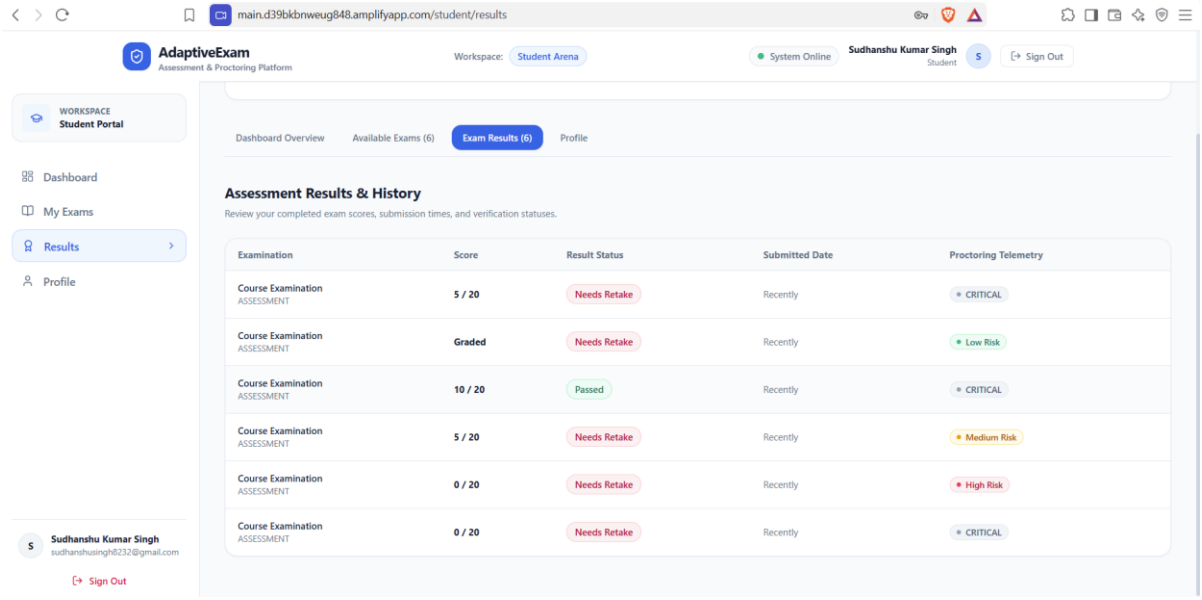
A.8.7 — Student Arena screens



A.8.8 — Student Arena screens



A.8.9 — Student Arena screens



A.8.10 — Student Arena screens